



Human Visual System

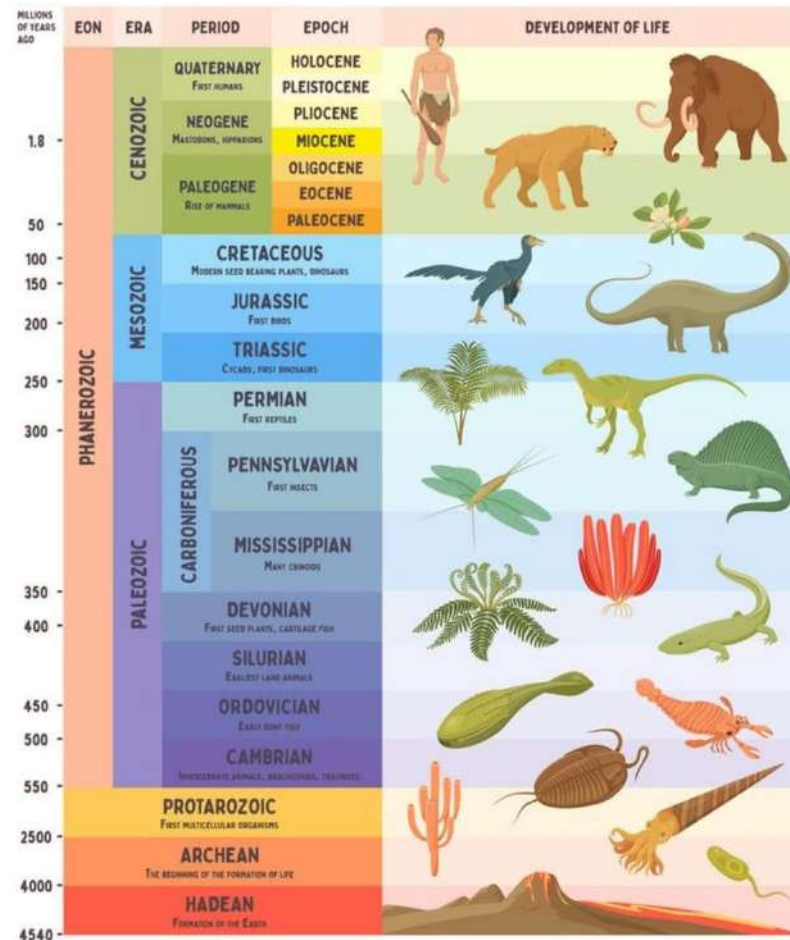
Taras Holotyak

Content

- Introduction: biological imaging systems
- Human eye as primary component of human visual system
- Primary visual cortex
- Applications

Human imaging systems

Geological timeline



Human imaging systems

Biological imaging systems: sea creatures



Nautilus belauensis
(450 million years)



Octopus
(200 million years)

Human imaging systems

Biological imaging systems: terrestrial creatures



Reptiles
(200 million years)



Spiders
(130 million years)

Human imaging systems

Biological imaging systems: terrestrial creatures



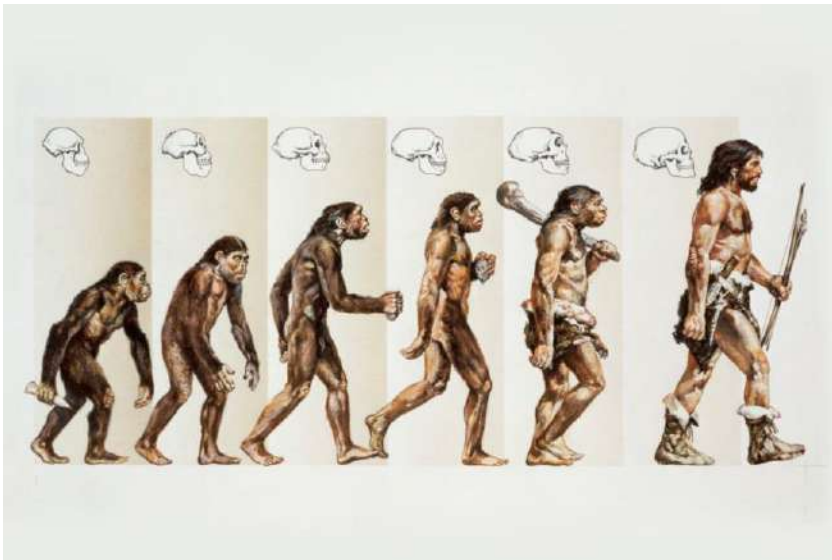
Birds
(155 million years)



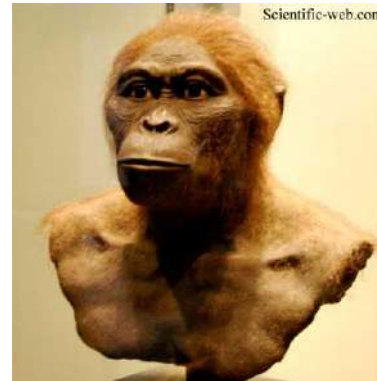
Mammals
(125 million years)

Human imaging systems

Australopithecus afarensis – Homo habilis – Homo erectus – Homo meanderthalensis - Homo sapiens

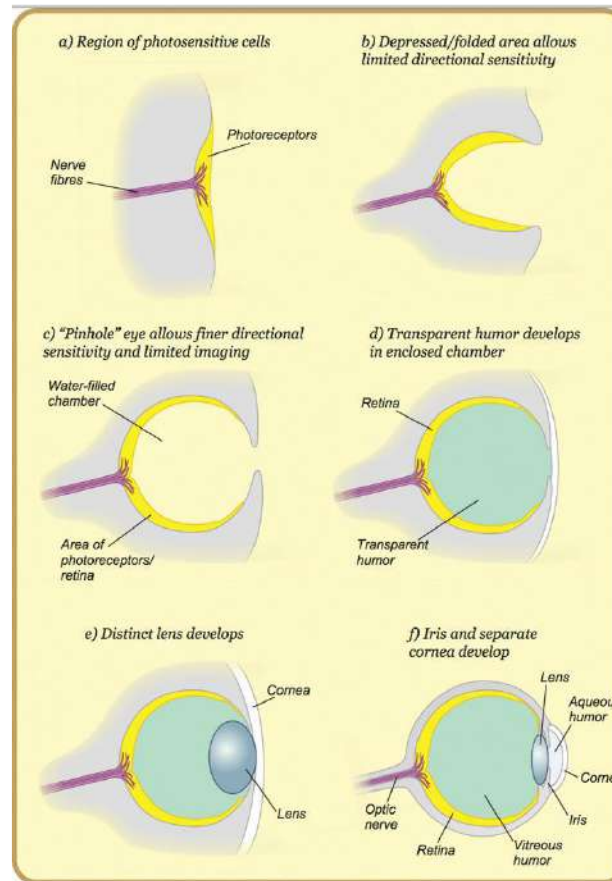


(3 million years)



Human imaging systems

Evolution of the eye



Human visual system

Interpretation of HVS in different historical epochs

Extromissive theories of vision

- Plato (350 B.C.) - from our eyes flows a light similar to the light of the sun
- “Therefore, when these three conditions concur, sight occurs, and the cause of sight is threefold: the light of the innate heat passing through the eyes, which is the principal cause, the exterior light kindred to our own light, which both acts and assists, and the light that flows from visible bodies, flame or color; without these the proposed effect [vision] cannot occur.” [Chalcidius, middle ages].
- Other non-material theories (spiritual, the “evil eye”)

Human visual system

Interpretation of HVS in different historical epochs

Extromissive theories faced many difficulties

- why do we see faraway objects instantaneously when we open our eyes?
- why do not the vision systems of different people looking at the same object interfere with each other?
- what if the eyes are closed when the visual spirit returns?
- objects create “material images” that are transported through the atmosphere and enter the eye (Aristotle 330 B.C.), but how do the material images of large objects enter the eye?

Human visual system

Interpretation of HVS in different historical epochs

Abu Ali al-Hassan ibn al-Hasan ibn al-Haytham (Alhazen) (1040)

– greatest optical scientist of the middle ages

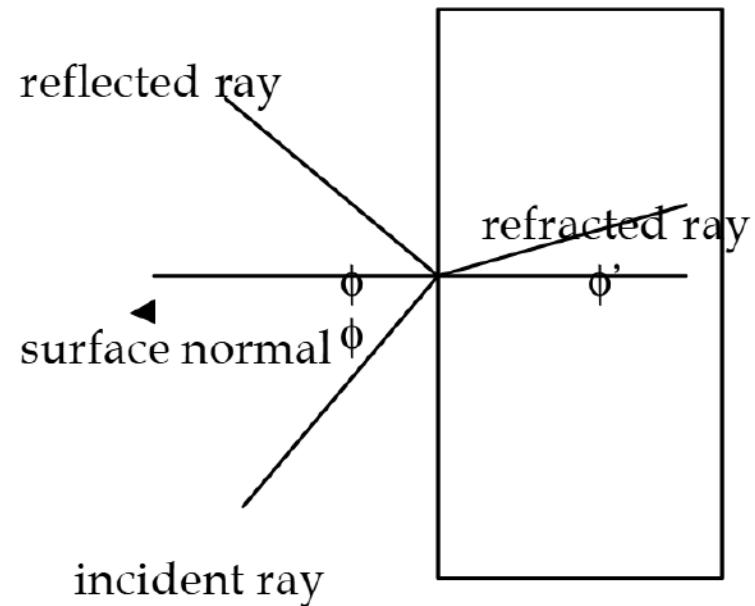
- Self luminous bodies: sun, moon, light
- Lights travel in straight lines
- When light hits an object, it irradiates every place
- Concept of medium: transparent and opaque
- Pointillist theory of vision - we see a collection of points on the surfaces of objects
- Geometric theory to explain the one-to-one correspondence between the world and the image formed in our eyes

Human visual system

Interpretation of HVS in different historical epochs

Image formation

- Ray of light leaves the light source, and travels along a straight line
- Light hits an object and is
 - reflected and/or
 - refracted
- If the object is our lens, then the useful light for imaging is the refracted light



Human visual system

Interpretation of HVS in different historical epochs

Ptolemy, Alhazen and refraction

- The phenomena of refraction was known to Ptolemy (150 B.C.)
- Alhazen's problem - since light from a surface point reaches the entire surface of the eye, how is it that we see only a single image of a point?
 - he assumed that only the ray that enters perpendicular to the eye effects vision
 - the other rays are more refracted, and therefore “weakened”
 - but in fact, the optical properties of the lens combine all of these rays into a single “focused” point under favorable conditions

Human visual system

Interpretation of HVS in different historical epochs

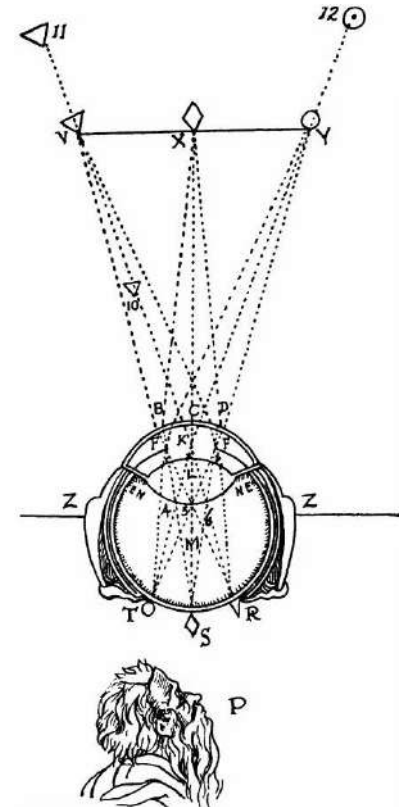
Johannes Kepler (1571-1630)

Founder of modern theories about optics and light

- Light has the property of flowing or being emitted by its source towards a distance place
- From any point the flow of light takes place according to an infinite number of straight line
- Light itself is capable of advancing to the infinite
- The lines of these emissions are straight and are called rays

Kepler's retinal theory

Even though light rays from “many” surface points hit the same point on the lens, they approach the lens from different directions. Therefore, they are refracted in different directions, separated by the lens.



Human visual system

Interpretation of HVS in different historical epochs

Modern theories of vision

Three main streams contribute to our understanding of vision:

- Psychology of perception: functionalities
- Neurophysiology: explanations
- Computational vision: more problems

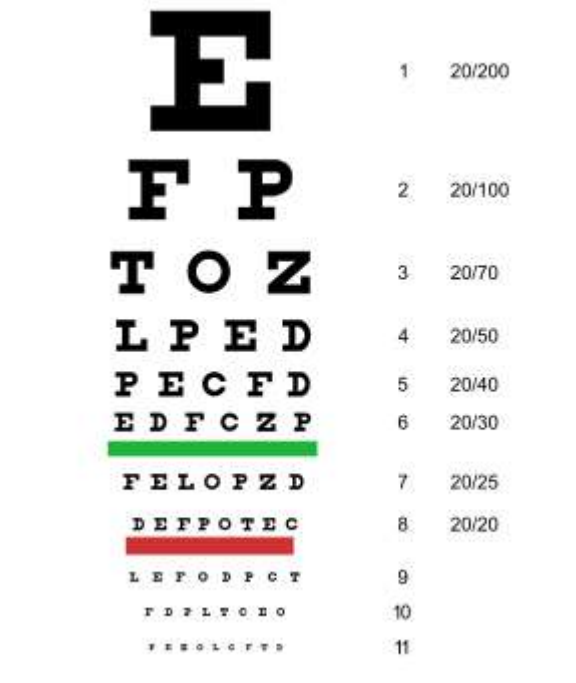
Modern partition of the human vision:

- Low-level (early) vision: parallelism, multiplexing, partitioning
- High-level vision: modularity.

Human visual system

Summary of human visual system (HVS)

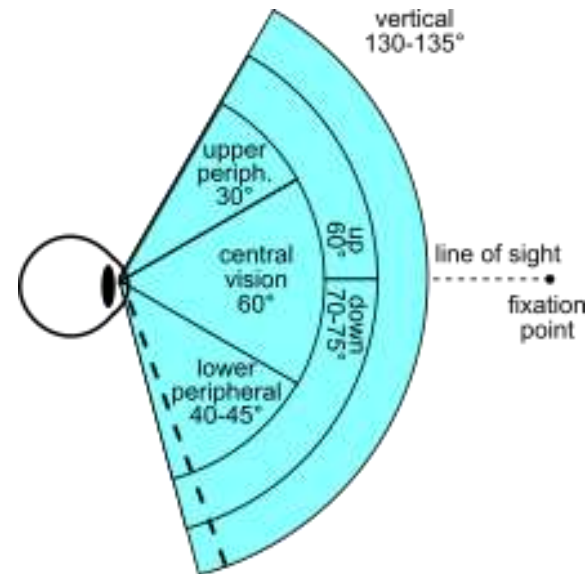
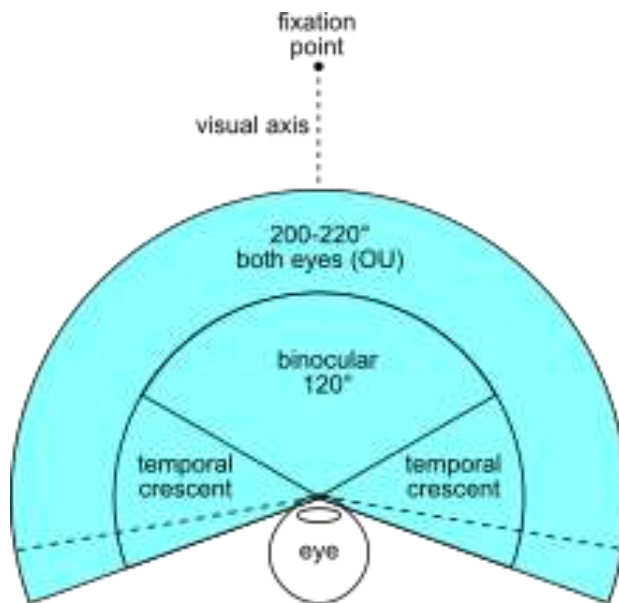
- visual acuity: 20/20 is ~ 1 arc min



Human visual system

Summary of human visual system (HVS)

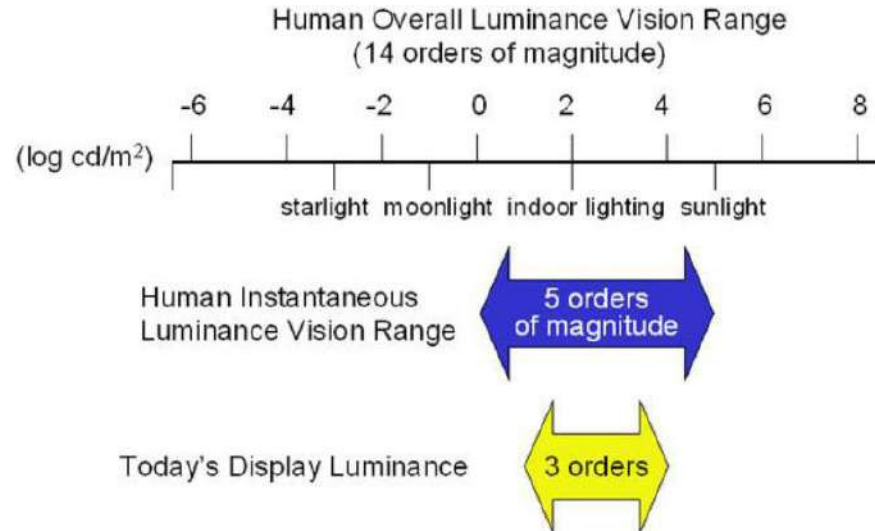
- field of view:
 - horizontal $\sim 200^\circ$ ($\sim 120^\circ$ binocular + $\sim 40^\circ$ monocular of each eye),
 - vertical $\sim 135^\circ$



Human visual system

Summary of human visual system (HVS)

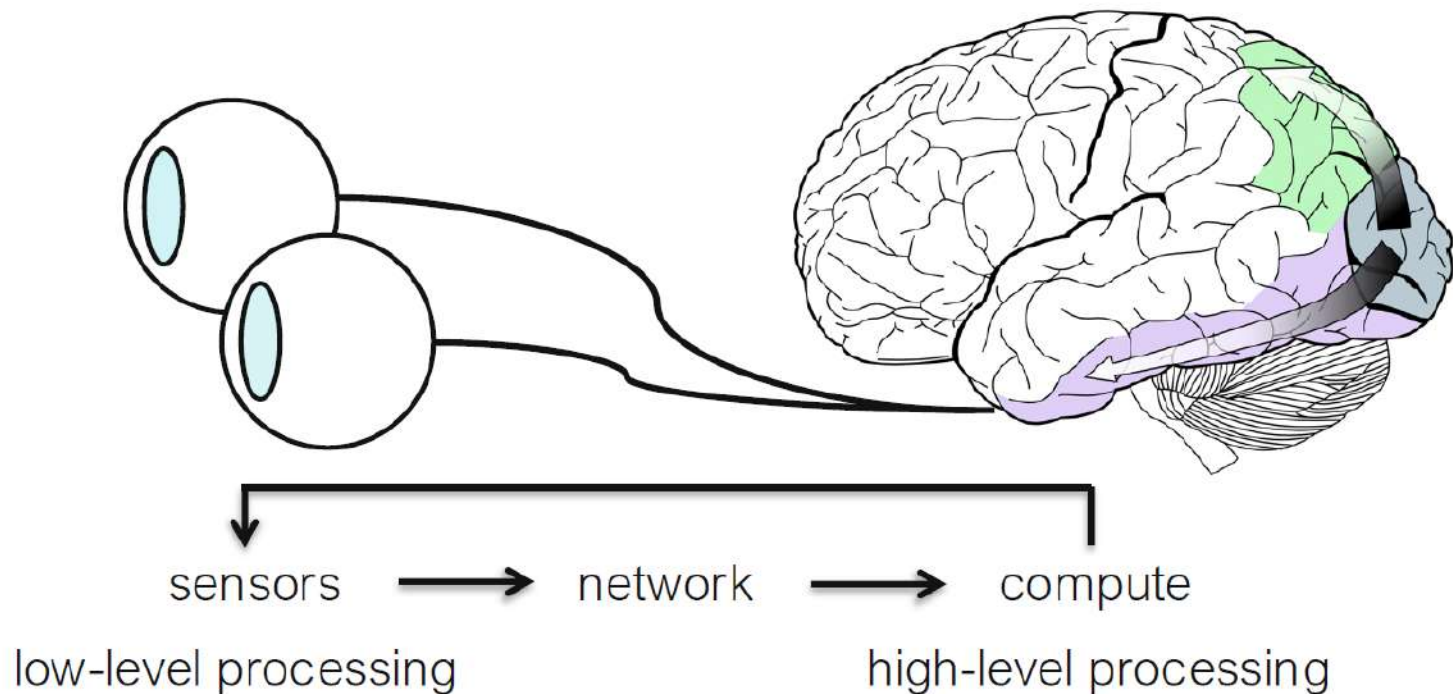
- temporal resolution: ~60 Hz (depends on contrast, luminance)
- dynamic range: instantaneous 6.5 f-stops, adapt to 46.5 f-stops



- color: everything in the CIE xy diagram; distances are linear in CIE Lab
- accommodation range: ~8cm to ∞ , degrades with age

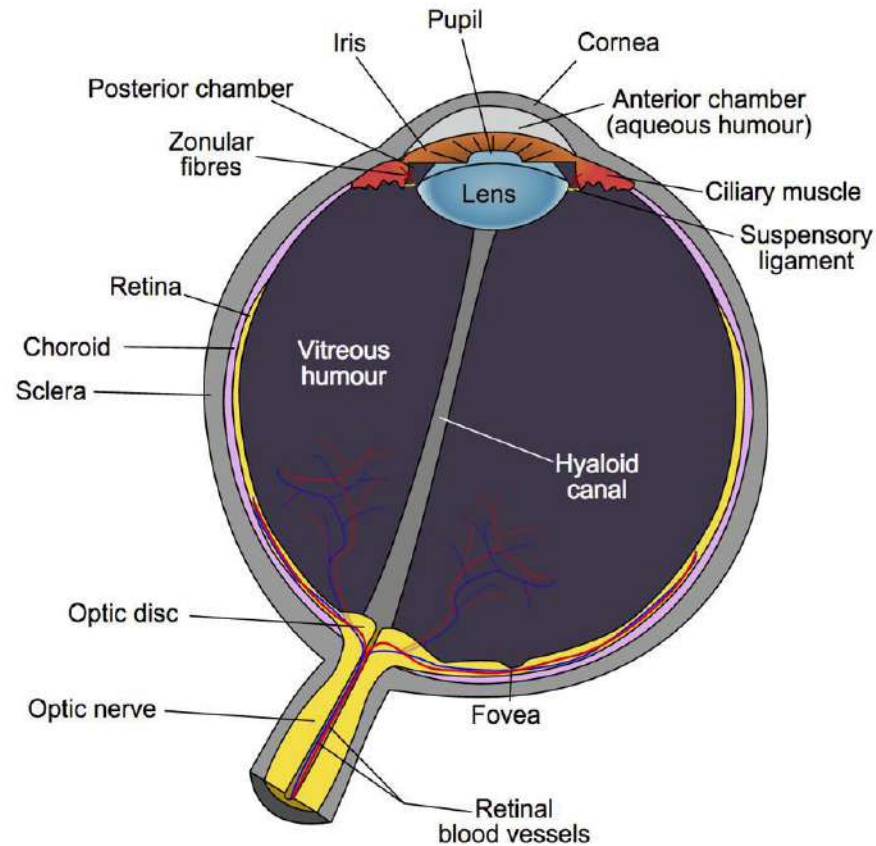
Human visual system

Overview of human visual system



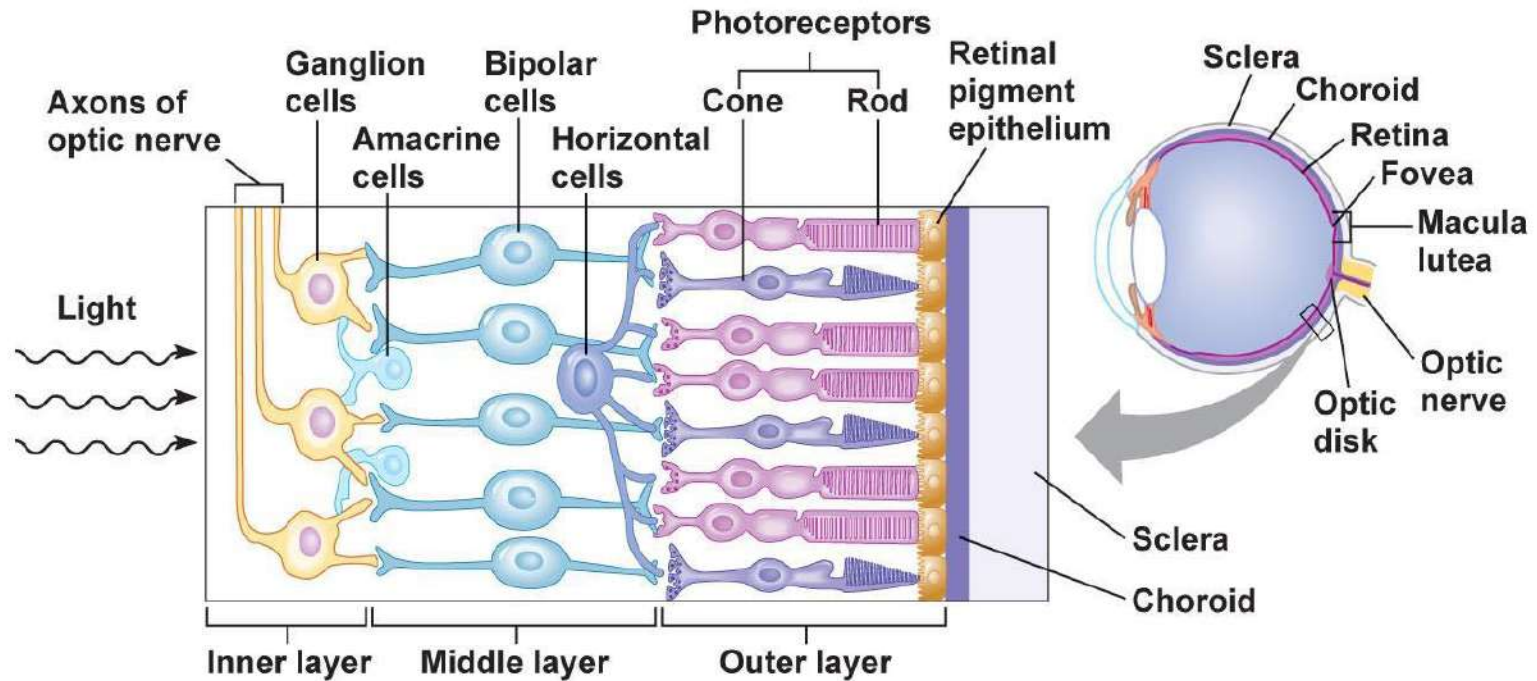
Human visual system

Anatomy of the human eye



Human visual system

Retina



© 2011 Pearson Education, Inc.

Three layers of cells:

- Receptor cells
- Collector cells
- Retinal ganglion cells

Human visual system

Retina

- Trade off: sensitivity to light vs. spatial resolution
- Two parallel systems:
 - One that is dedicated to sensitivity to light (Rods):
 - high sensitivity (sensitive to low levels of light: scotopic light levels)
 - extensive convergence onto collector & ganglion cells

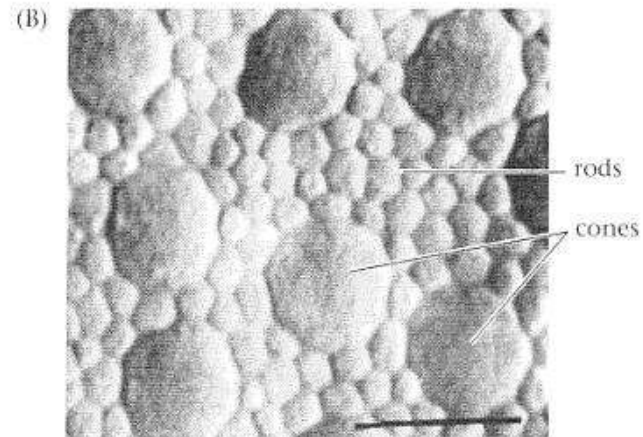
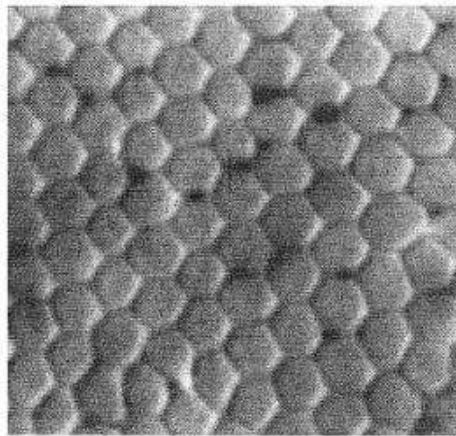
⇒ Low resolution image of the world that persists even in low illumination condition
 - One that is dedicated to spatial resolution (Cons):
 - sensitive to higher levels of light: photopic light levels
 - much limited convergence

⇒ High resolution image of the world in good illumination.

Human visual system

Photoreceptor mosaics

Rods and cones can be considered as eye's pixels



Human visual system

Photoreceptor mosaics

Cones (10^7) and rods (10^8)

Cones – color; rods – grey level

photopic scotopic

Cones come in three types

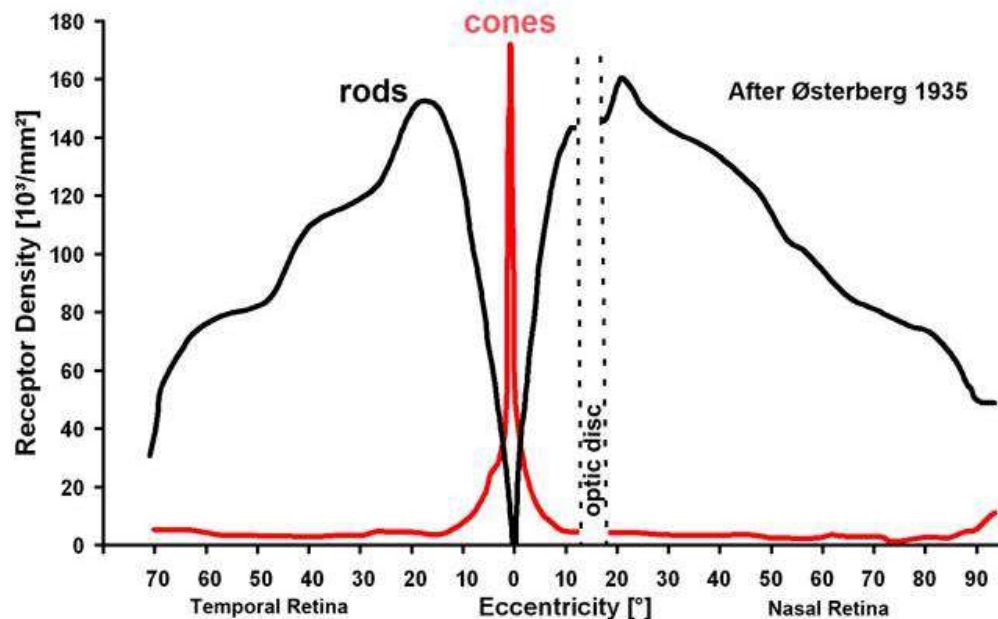
1. S – short wavelength (blue)
2. M – medium wavelength (green)
3. L – long wavelength (red)

Insufficient bandwidth of optic nerve implies coding

Human visual system

Photoreceptor mosaics

Rod and cone densities

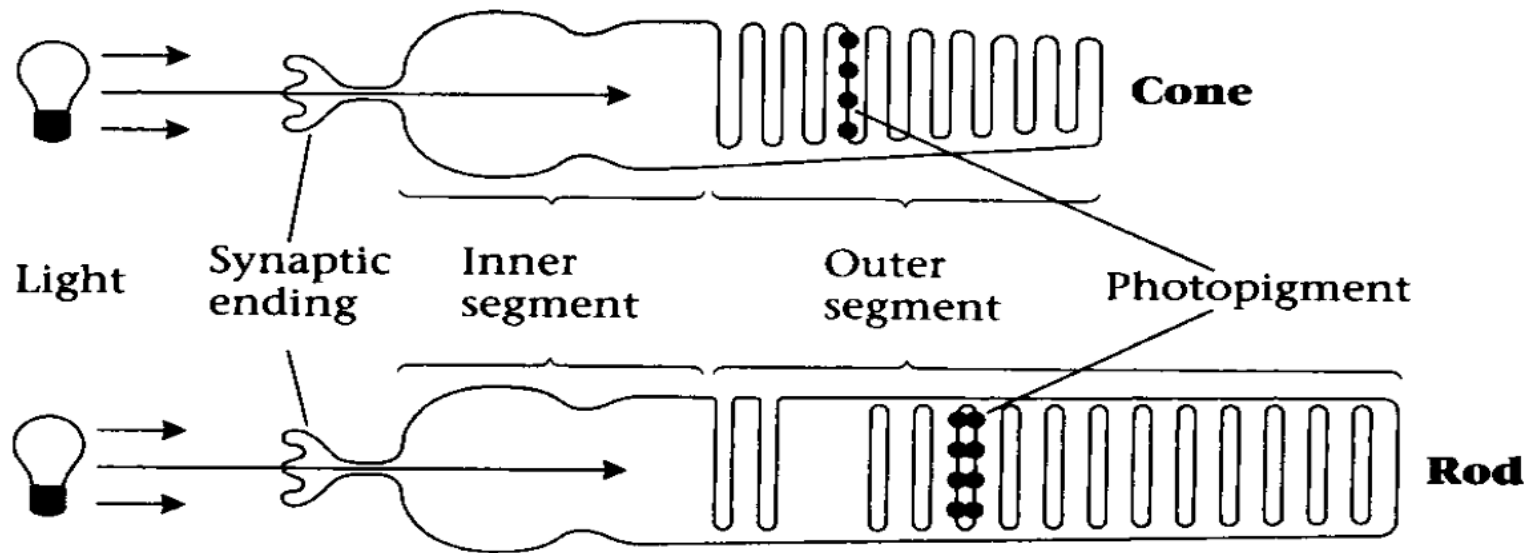


- No sensors on blind spot
- Most cones on fovea
- Rods elsewhere

Human visual system

Photoreceptor mosaics

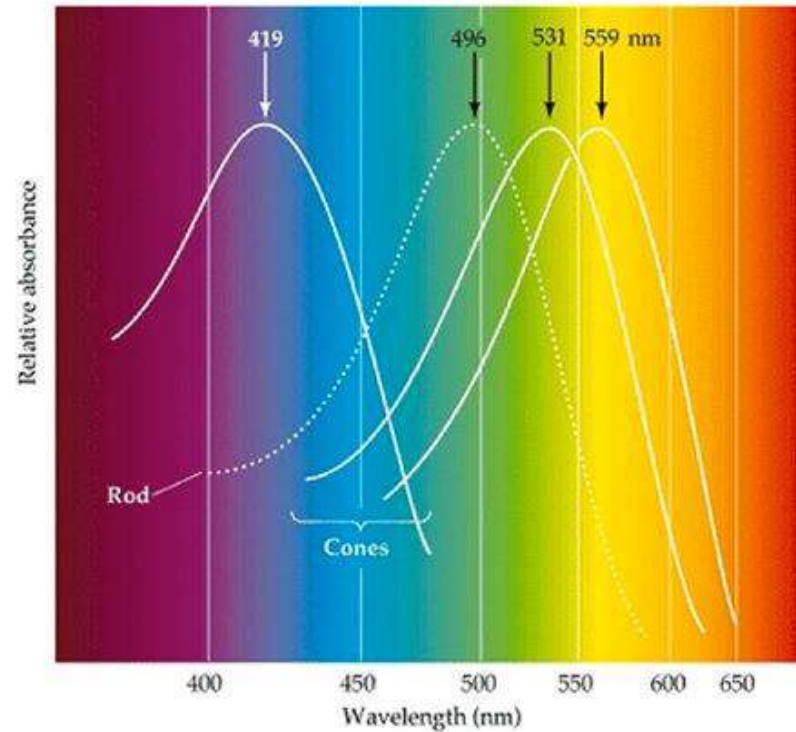
After dark adaptation, a single rod can respond to a single photon



Human visual system

Photoreceptor mosaics

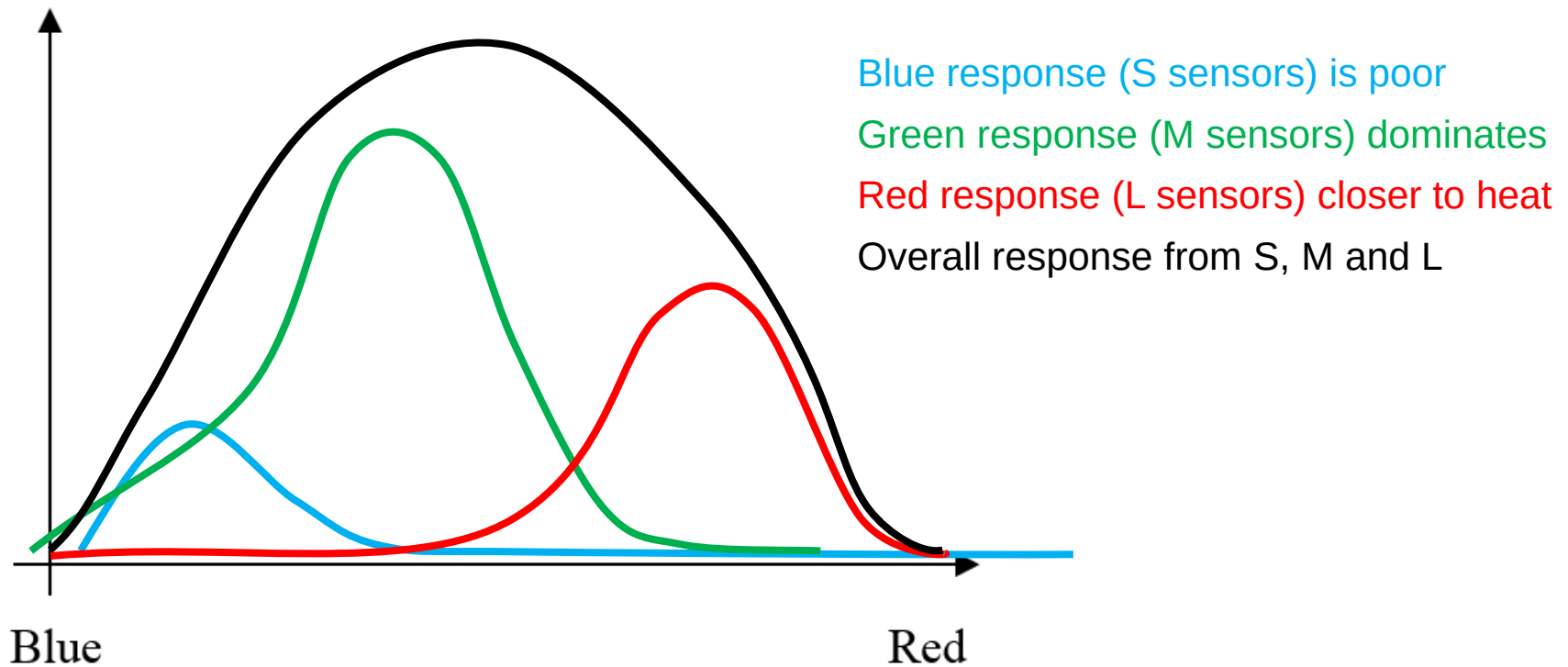
Color perception



Human visual system

Photoreceptor mosaics

Spectral response



Human visual system

Photoreceptor mosaics

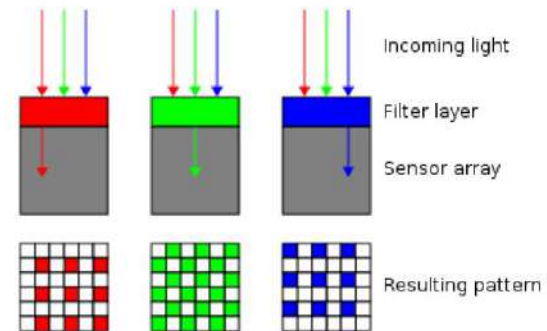
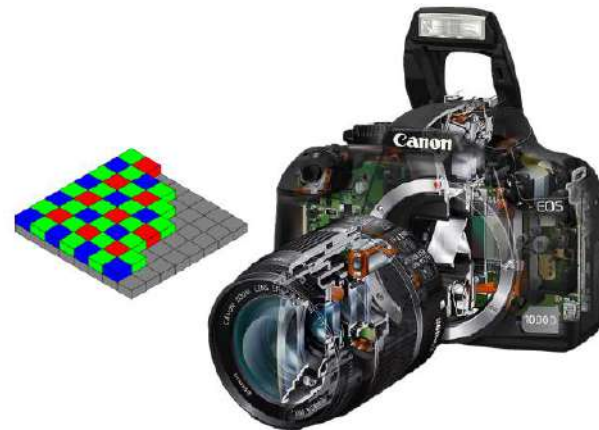
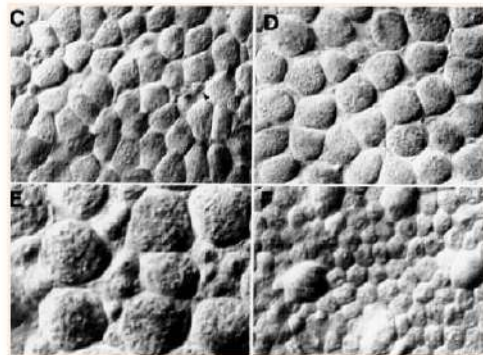
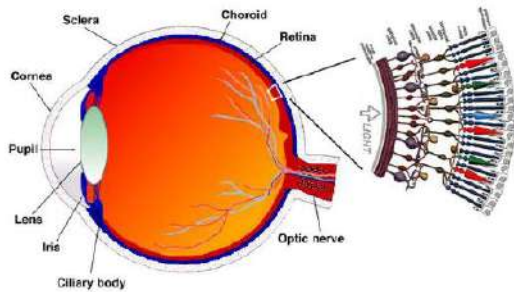
Limitation of human vision:

- blood vessels and other cells in front of photoreceptors
- shadows cast on photoreceptors
- non-uniform brightness
- the image is upside-down!
- high resolution vision only in the fovea
 - only one small fovea in man
 - other animals (birds, cheetas) have different foveal organizations
- blind spot

Human visual system

Photoreceptor mosaics

Eye vs. camera



Human visual system

Retinal ganglion cells

- First stage of visual processing
- Function: Absolute levels of illumination is replaced by a retinotopic map of “differences”
- How: center-surrounding organization of their receptive fields:
 - on-center (off-surrounding) cells
 - off-center (on-surrounding) cells
- Why: objects are not associated with any particular brightness, but with differences in brightness between themselves and the background
- The differences can be amplified without having to represent the enormous range of values that would result from the amplification of absolute values
⇒ groundwork for perception of objects

Human visual system

Retinal ganglion cells

Another partition:

- M and P cells:
- Feeds into the M and P channels (magnocellular and parvocellular layers in LGN)
- Trade-off: temporal vs. spatial resolution
- M cells: input from large number of photoreceptors $\Rightarrow$ good light sensitivity, good temporal resolution (can sample easily from large input), low spatial resolution.
- P cells: input from small number of photoreceptors $\Rightarrow$ good spatial resolution, poor temporal resolution.

Human visual system

Retinal ganglion cells

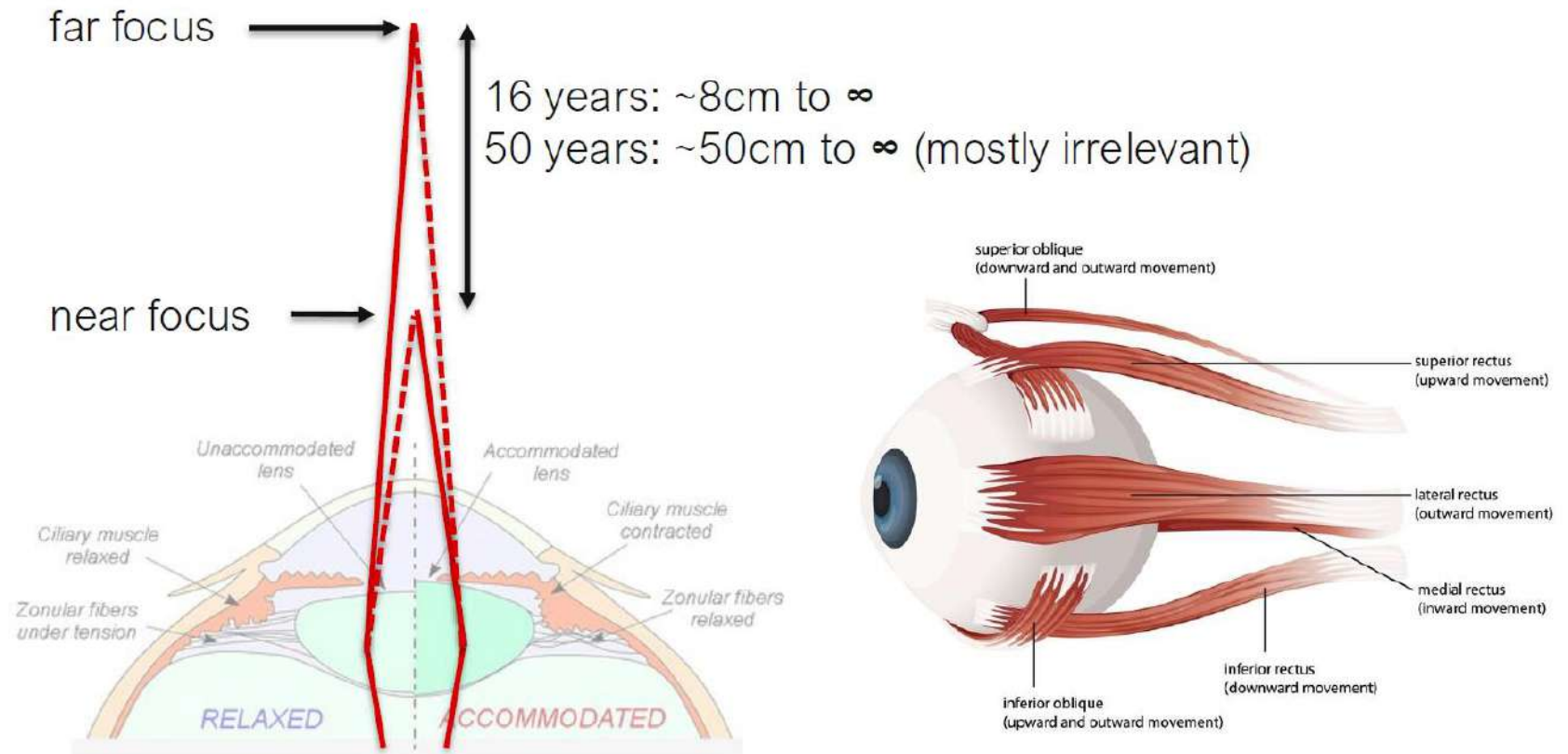
Another partition:

Trade-off: temporal vs. spatial resolution

- M cells are larger, faster nerve conduction velocities, responses are more transient
- P cells show color sensitivity, M cells do not.
- M cells: Temporal resolution $\Rightarrow$ motion perception, sudden stimulus
- P cells: Spatial resolution $\Rightarrow$ color, texture, patterns (major role in object perception).

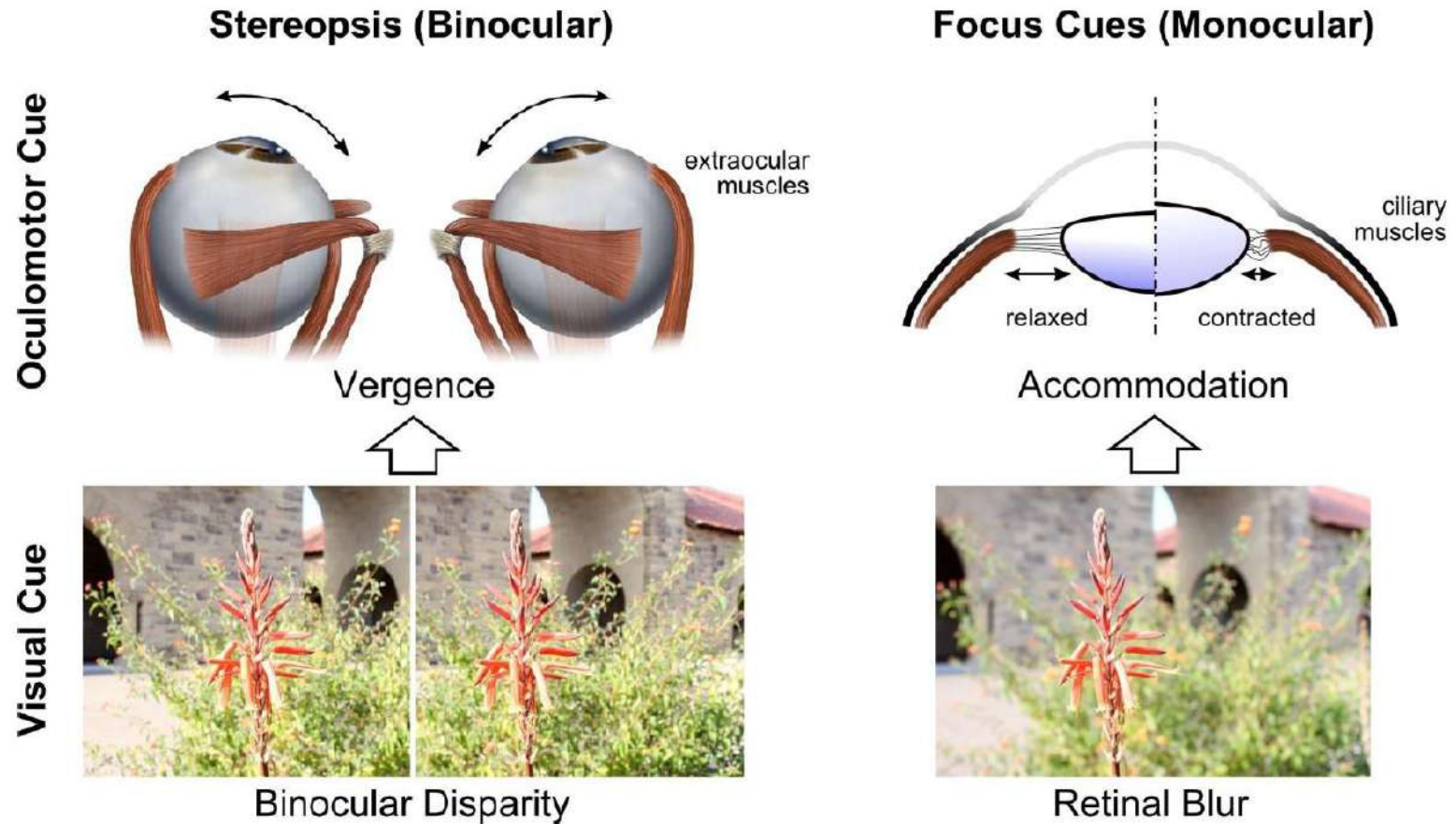
Human visual system

Oculomotor process



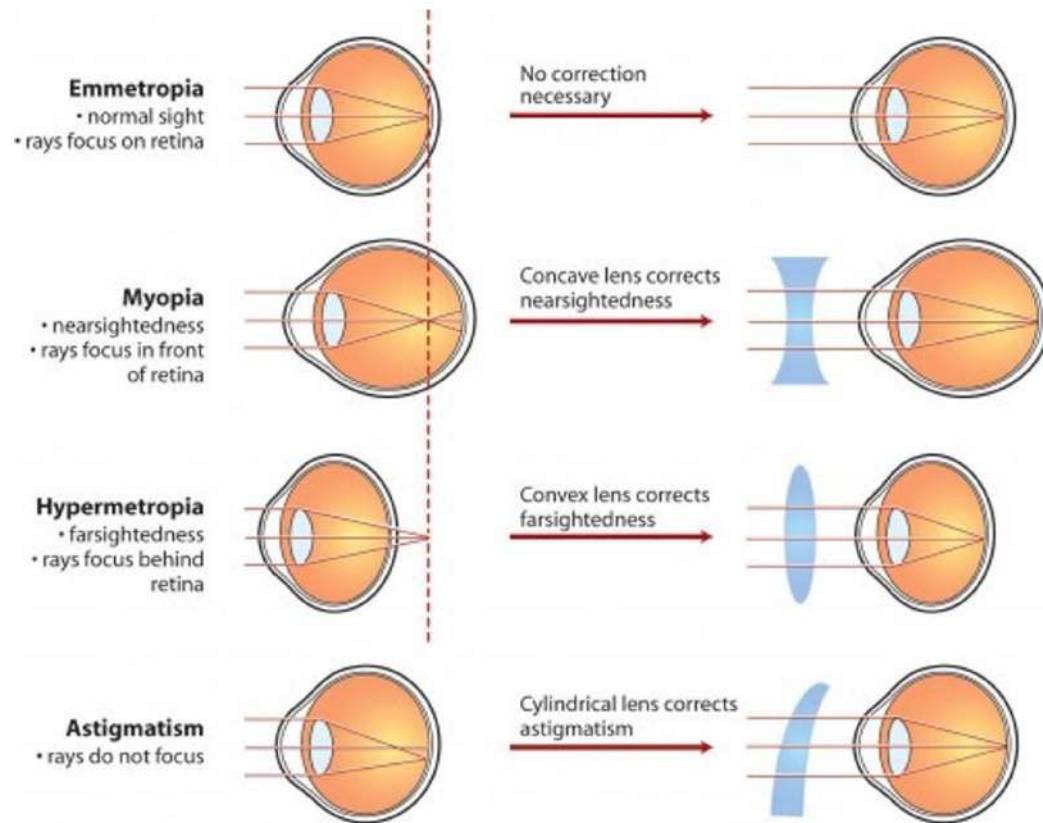
Human visual system

Oculomotor processes + Visual Cues



Human visual system

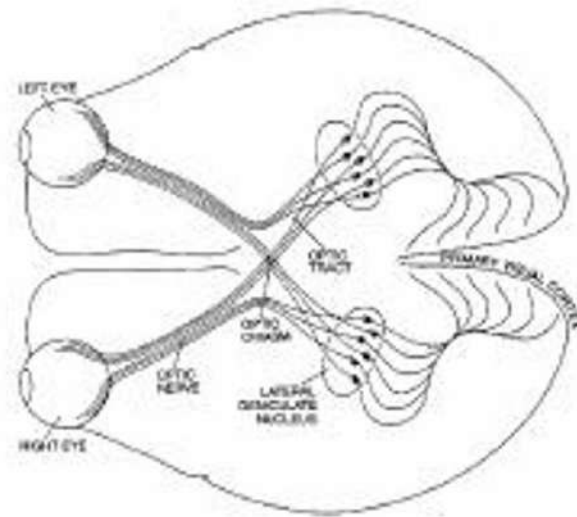
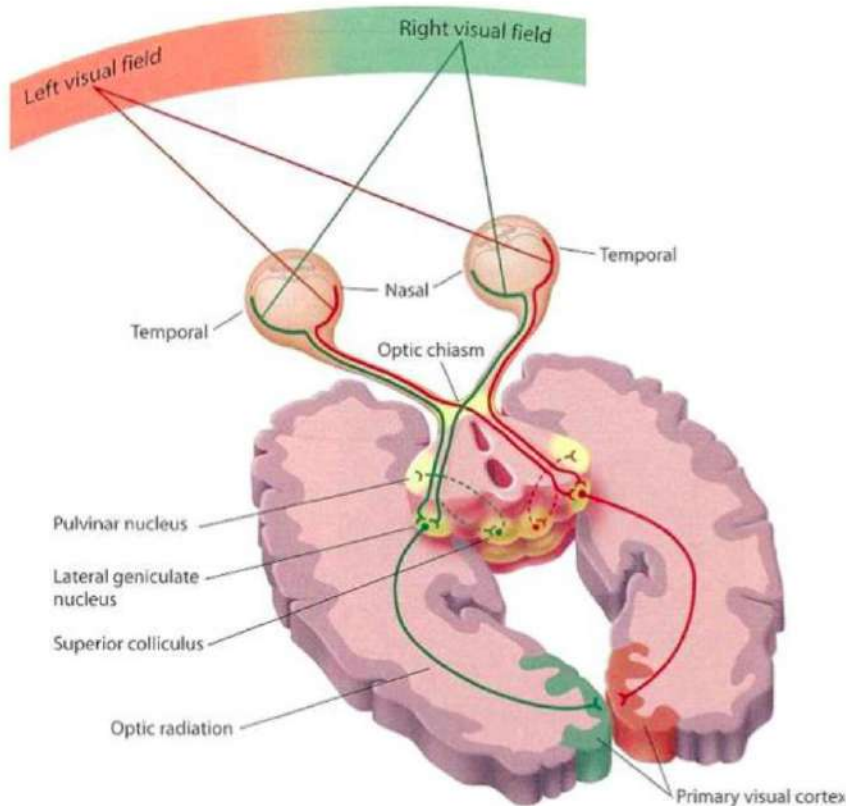
Refractive errors



Human visual system

Primary visual cortex

Anatomy of visual cortex pathway



Visual pathways

- Bundle of axons leaving the eye: optic nerve
- Split into a number of pathways

Human visual system

Primary visual cortex

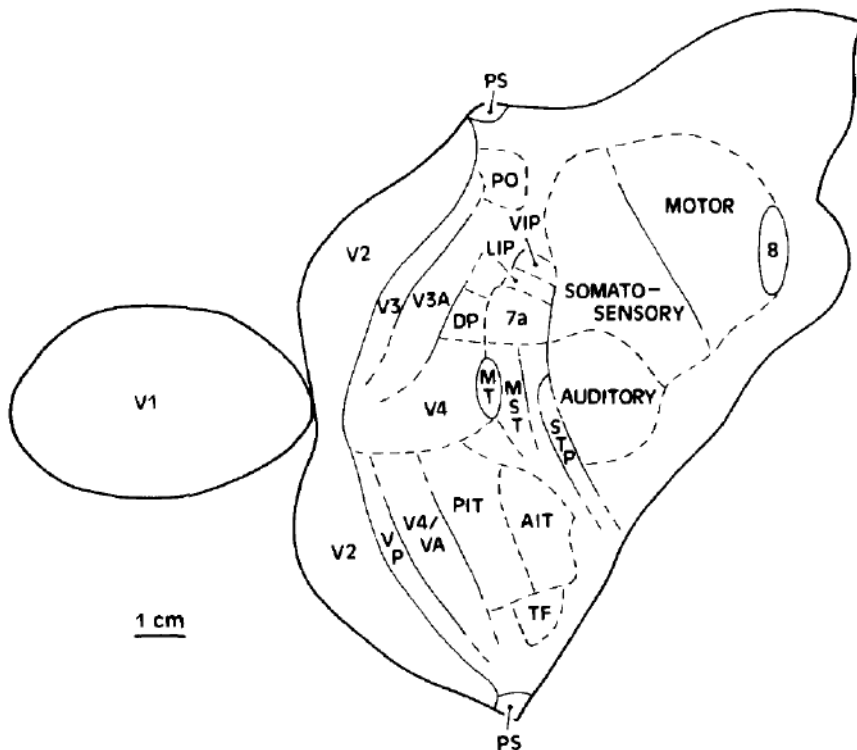
The lateral geniculate nucleus (LGN):

- One LGN in each cerebral hemisphere
- Magnocellular layers (two) : feed from M-cells
 - Best temporal resolution
- Parvocellular layers (four) : feed from P-cells
 - Best spatial resolution, wavelength sensitivity
- Another example of division of labor and multiplexing
- Neurons in all layers show center-surrounding organization
- Retinotopy in LGN and beyond: all layers keep retinotopic organization of the image

Human visual system

Primary visual cortex

Visual processing areas (anatomical map)



- David H. Hubel & Torsten N. Wiesel: Nobel prize
- Three types of cells (1962):
 - Center-surrounding
 - Simple cells:
 - like center-surrounding with elongated excitatory and inhibitory regions
 - edges at particular location and orientation
 - Complex cells:
 - more abstract type of visual information. Partially independent of location within the visual field

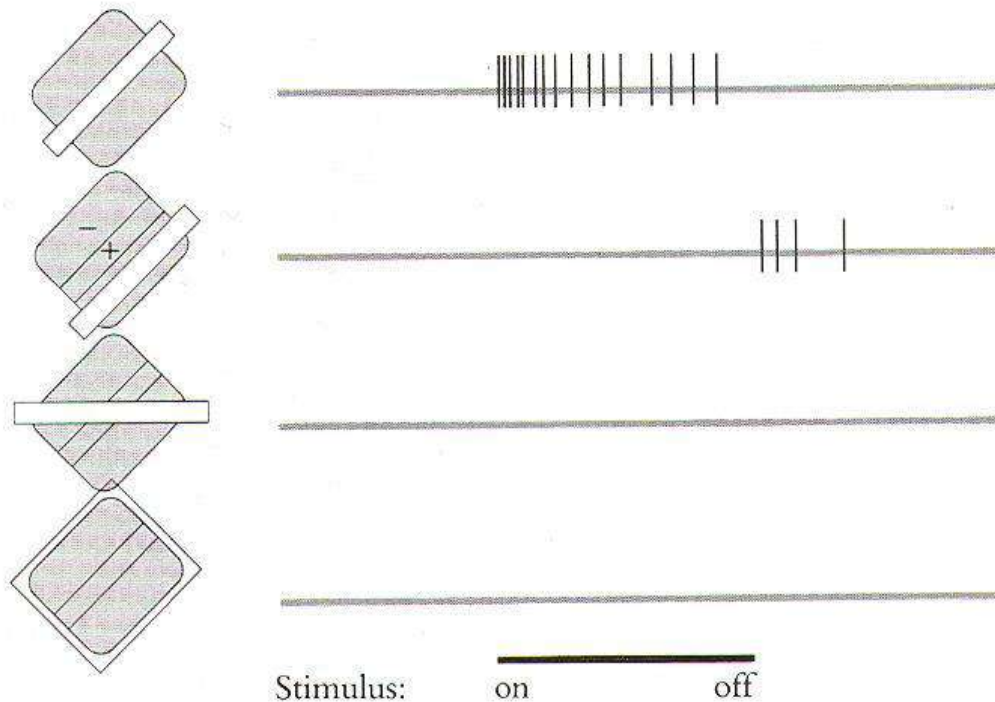
Visual processing areas (block-diagram)



Human visual system

Primary visual cortex

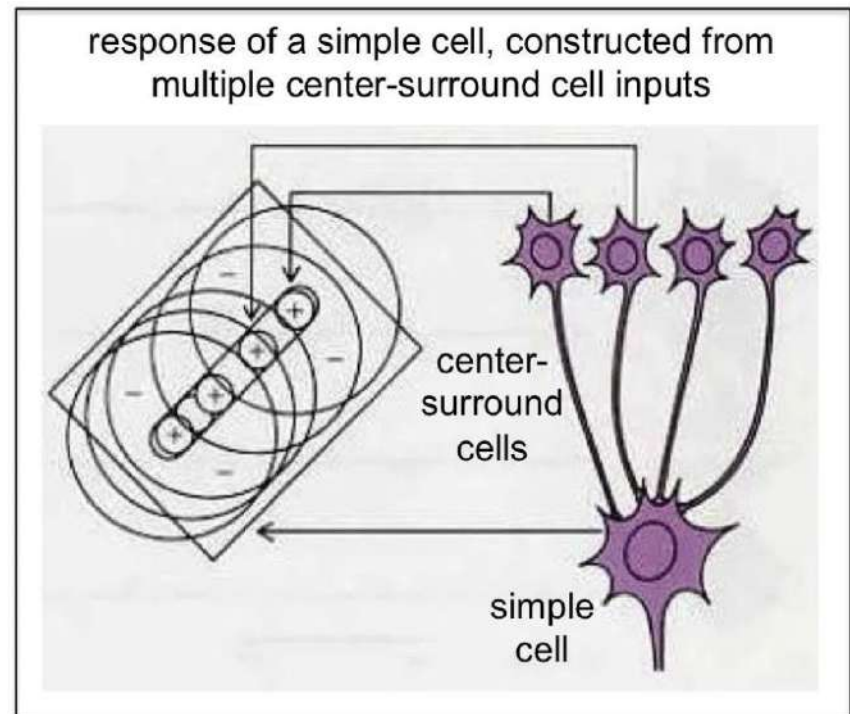
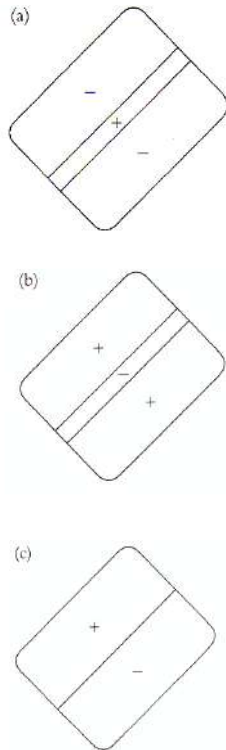
Orientation sensitivity of V1



Human visual system

Primary visual cortex

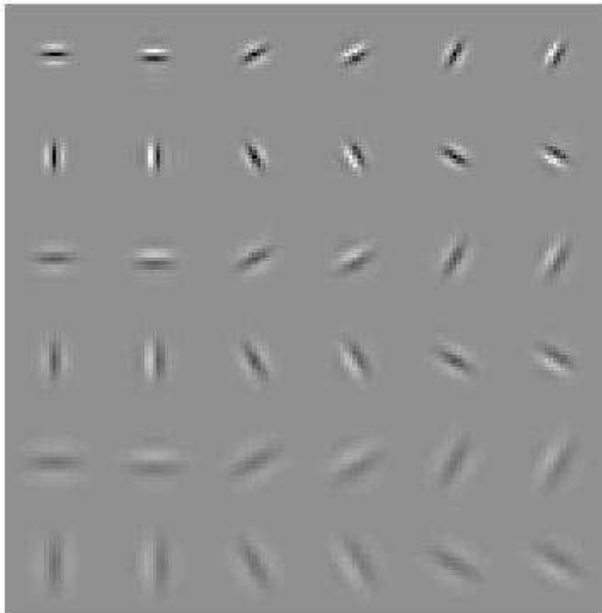
Receptive fields of simple cells



Human visual system

Primary visual cortex

Modelling simple cells



- Feed forward sequence or hierarchy of visual processing:

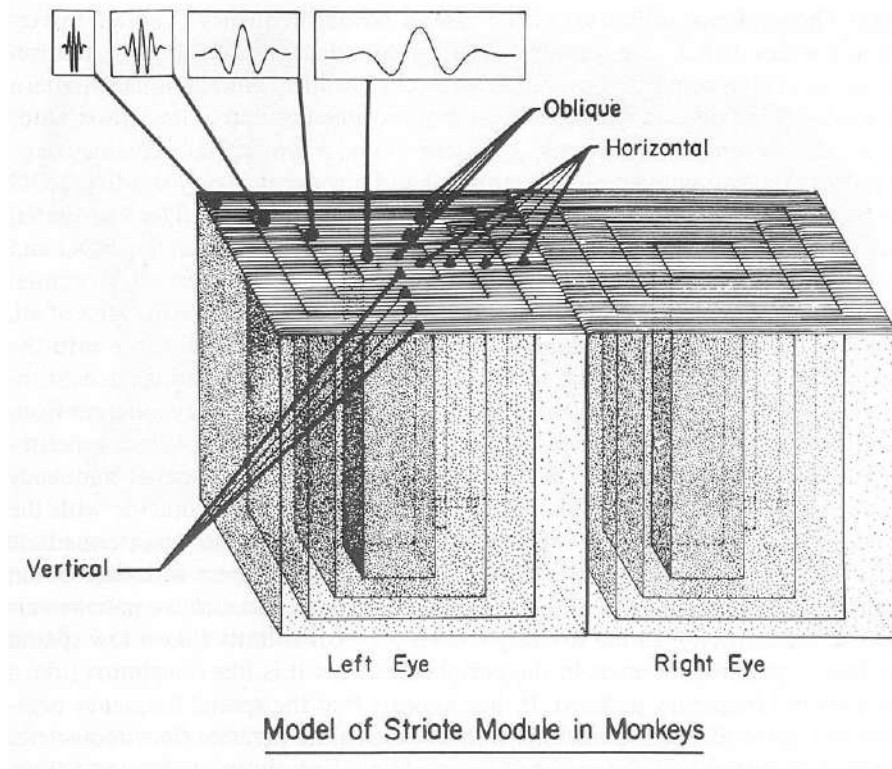
Center-surrounding → Simple → Complex

- Cells' responses become increasingly specific w.r.t the form of the stimulus (ex. oriented edges or bars)
- Increasingly general w.r.t viewing conditions (from just one location to a range of locations)
- These dual-trends are essential for object recognition
- Can respond to specific form (like familiar face) generalized over changes in size, orientation, view point)
- More recent research: lateral interaction plays important role (Gilbert 1992)

Human visual system

Primary visual cortex

Hypercolumns in visual cortex

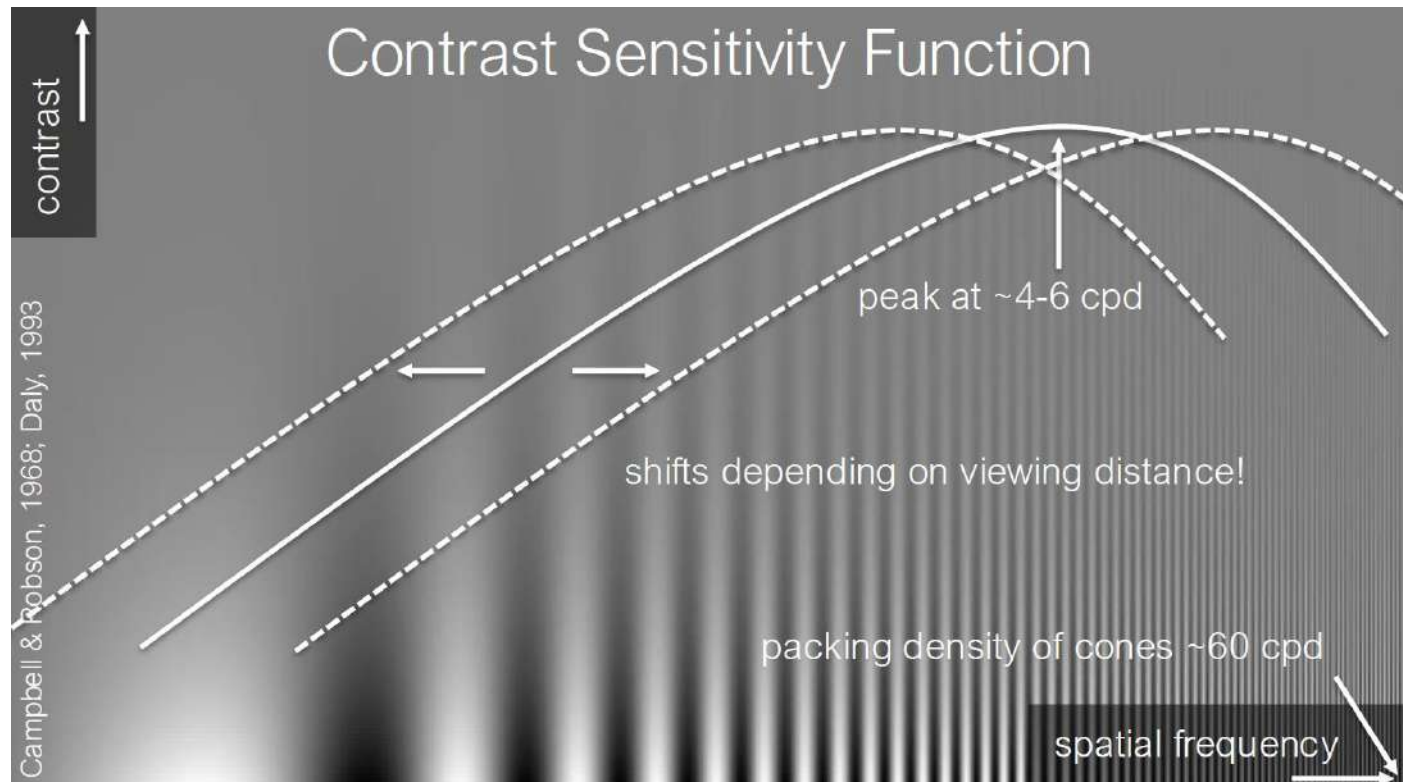


- Organization and orientation selectivity
- Spatial arrangement of cells for minimizing the distance between neurons representing similar stimulus along three different stimulus dimensions:
 - Eye of origin
 - Orientation
 - Retinotopic location
- Hebb rule: neurons that fire together wire together

Human visual system

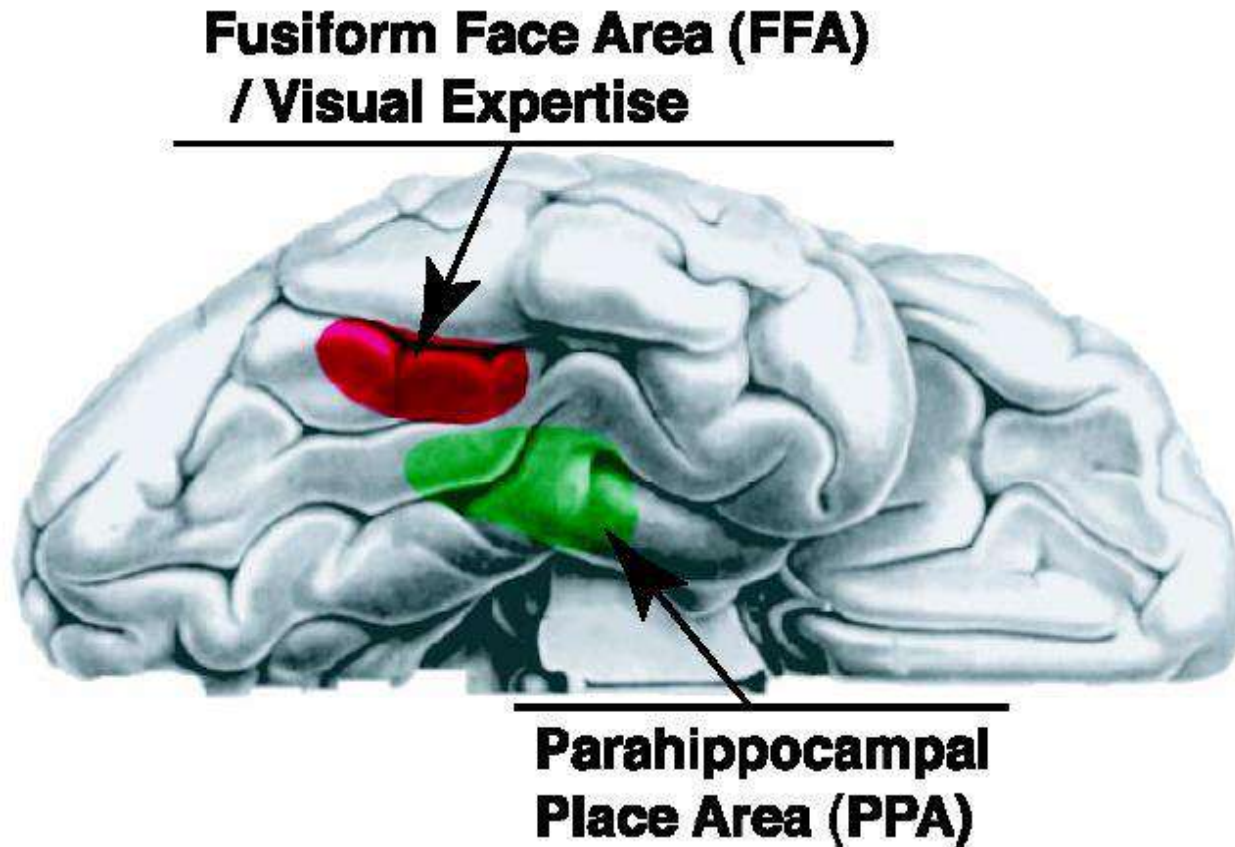
Primary visual cortex

Contrast effects



Human visual system

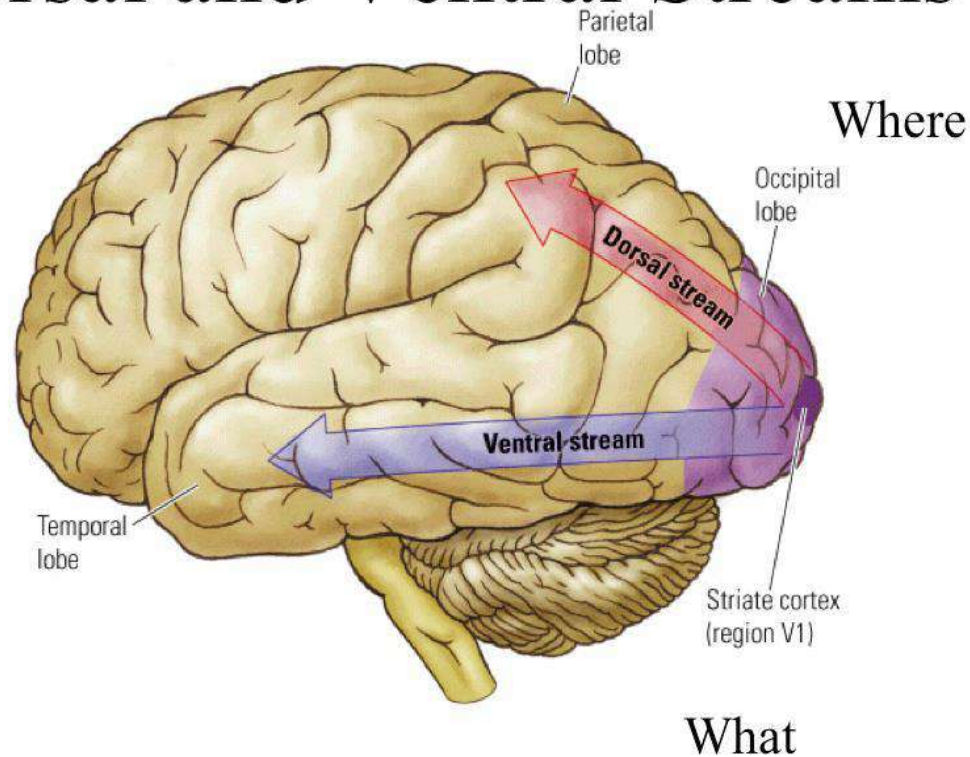
Primary visual cortex



Human visual system

Primary visual cortex

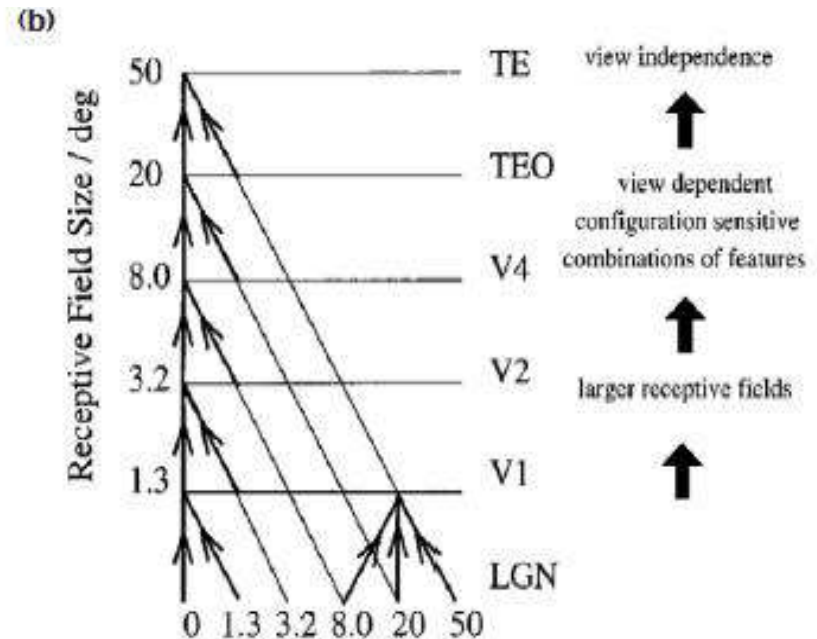
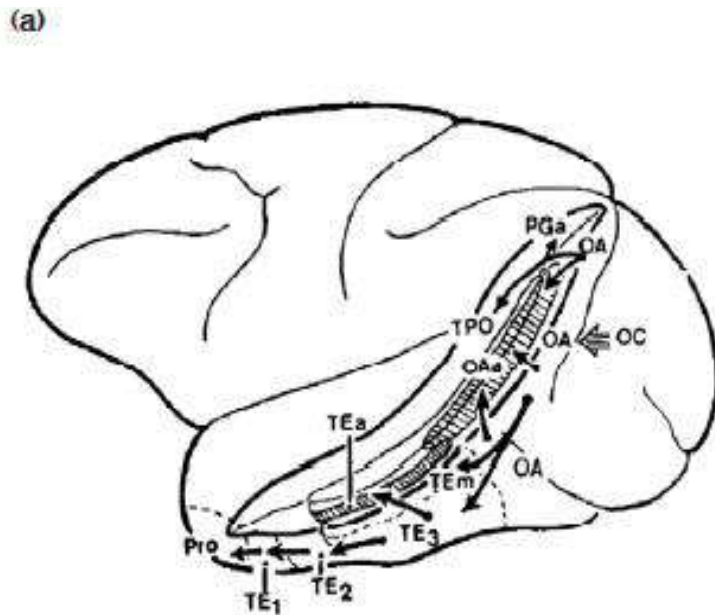
Dorsal and Ventral Streams



Human visual system

Primary visual cortex

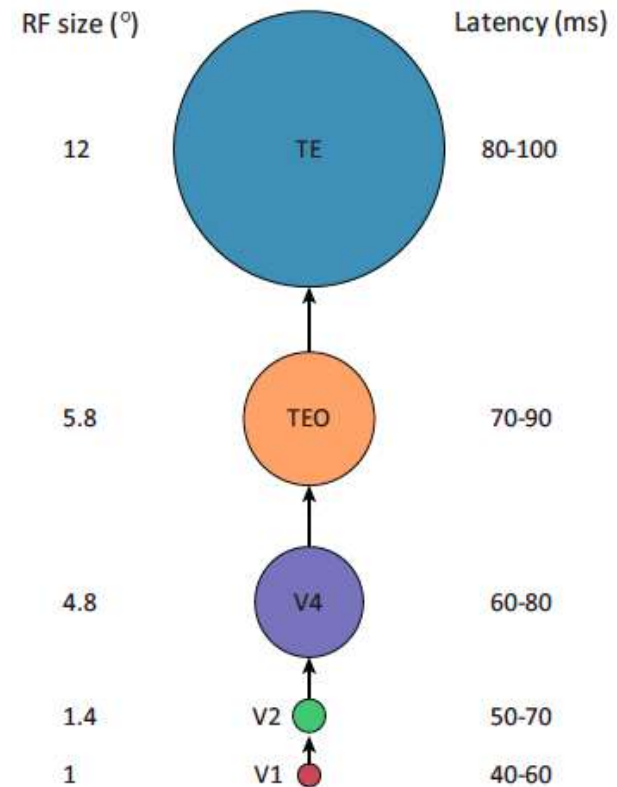
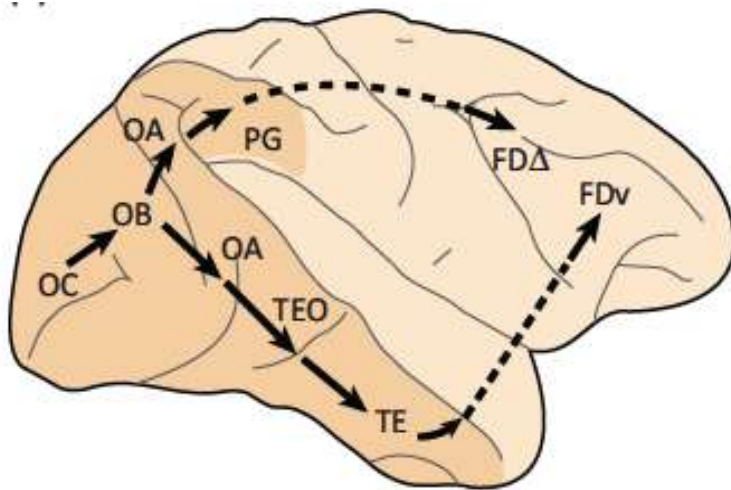
Model of ventral stream



Human visual system

Primary visual cortex

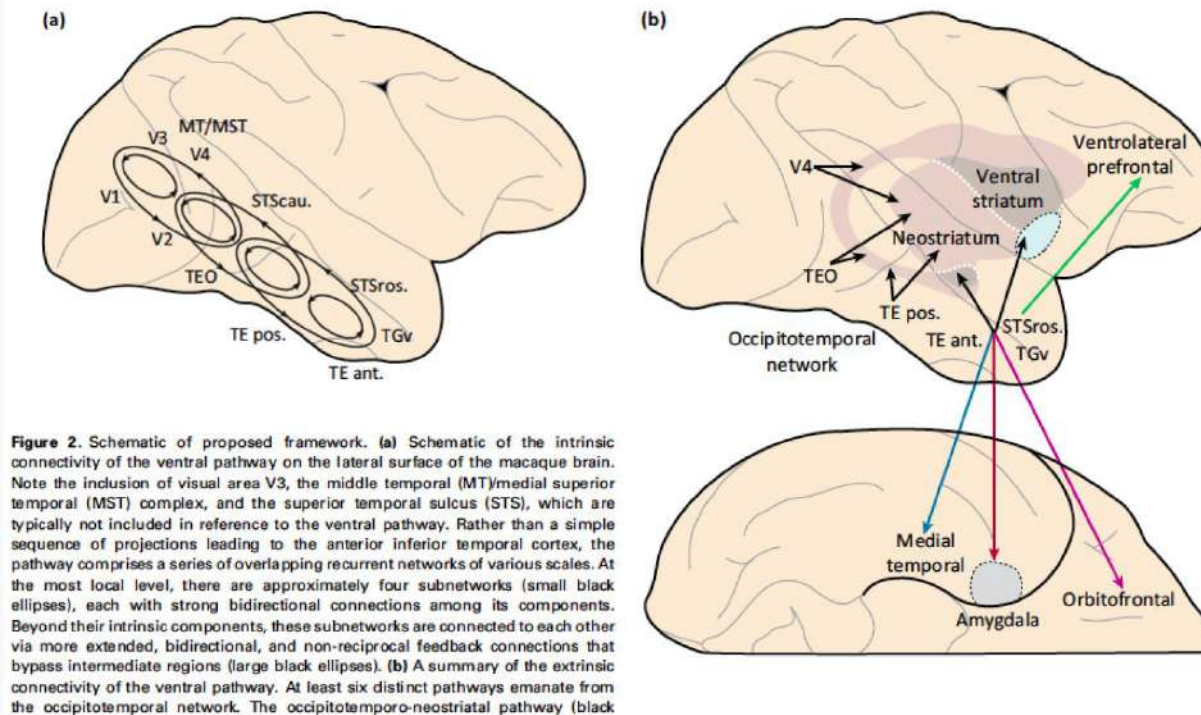
Model of ventral stream



Human visual system

Primary visual cortex

Intrinsic & Extrinsic Connectivity of the Ventral Stream (Kravitz, Saleem, Baker, Ungerleider, Mishkin, TICS, 2013)



Human visual system

Primary visual cortex

- Neurons show increasing specificity higher in the visual pathway
- V1 simple and complex cells are orientation-tuned
- Convolution with a linear kernel followed by simple non-linearities is a good model for computation in retina, LGN and V1, but beyond that we do not have satisfactory computational models
- Good designs of visual systems are likely to be hierarchical and “mostly” feedforward

Neuroscience & Computer Vision Features

- Orientation selectivity of simple and complex cells in V1 inspired features such as SIFT and HOG.
- A feed-forward view of processing in the ventral stream with layers of simple and complex cells led to the neocognitron and subsequently convolutional networks.
- Now it is known that the ventral stream is much more complicated with bidirectional as well as feedback connections. So far this has not been exploited much in computer vision.

Human visual system

Network models inspired by human visual systems

Neocognitron: A Self-organizing Neural Network Model for a Mechanism of Pattern Recognition Unaffected by Shift in Position

Kunihiko Fukushima

NHK Broadcasting Science Research Laboratories, Kinuta, Setagaya, Tokyo, Japan

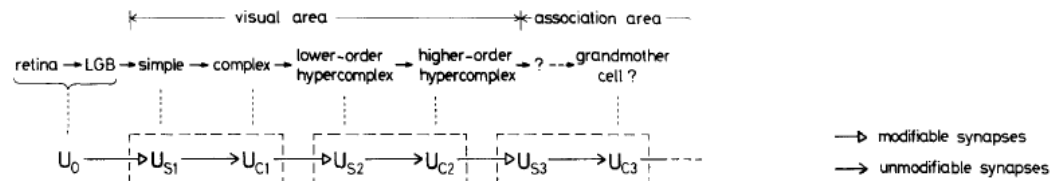


Fig. 1. Correspondence between the hierarchy model by Hubel and Wiesel, and the neural network of the neocognitron

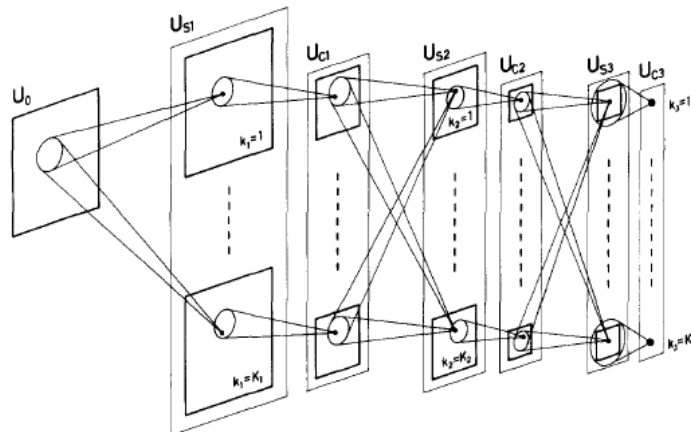


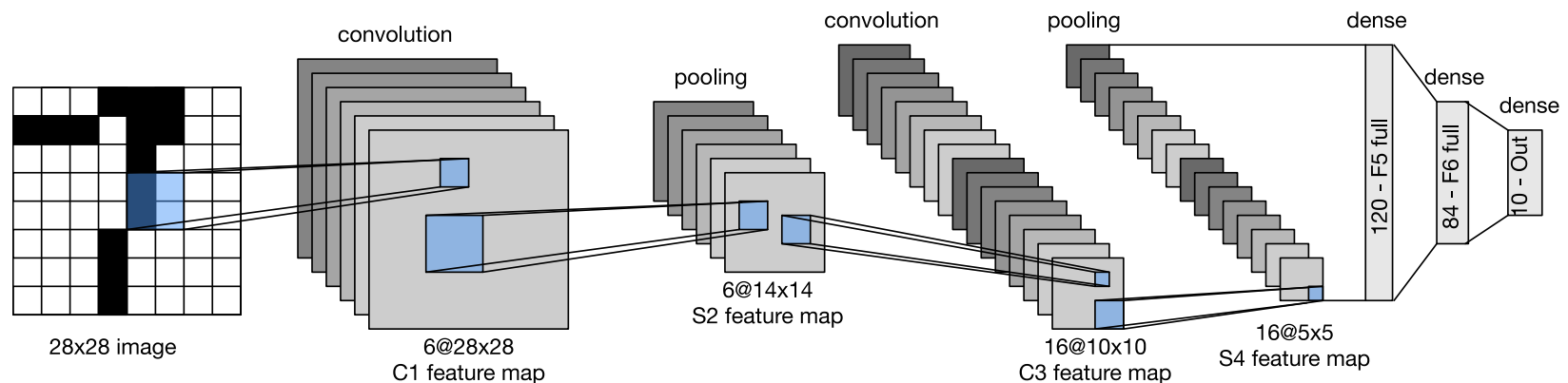
Fig. 2. Schematic diagram illustrating the interconnections between layers in the neocognitron

Human visual system

Network models inspired by human visual systems

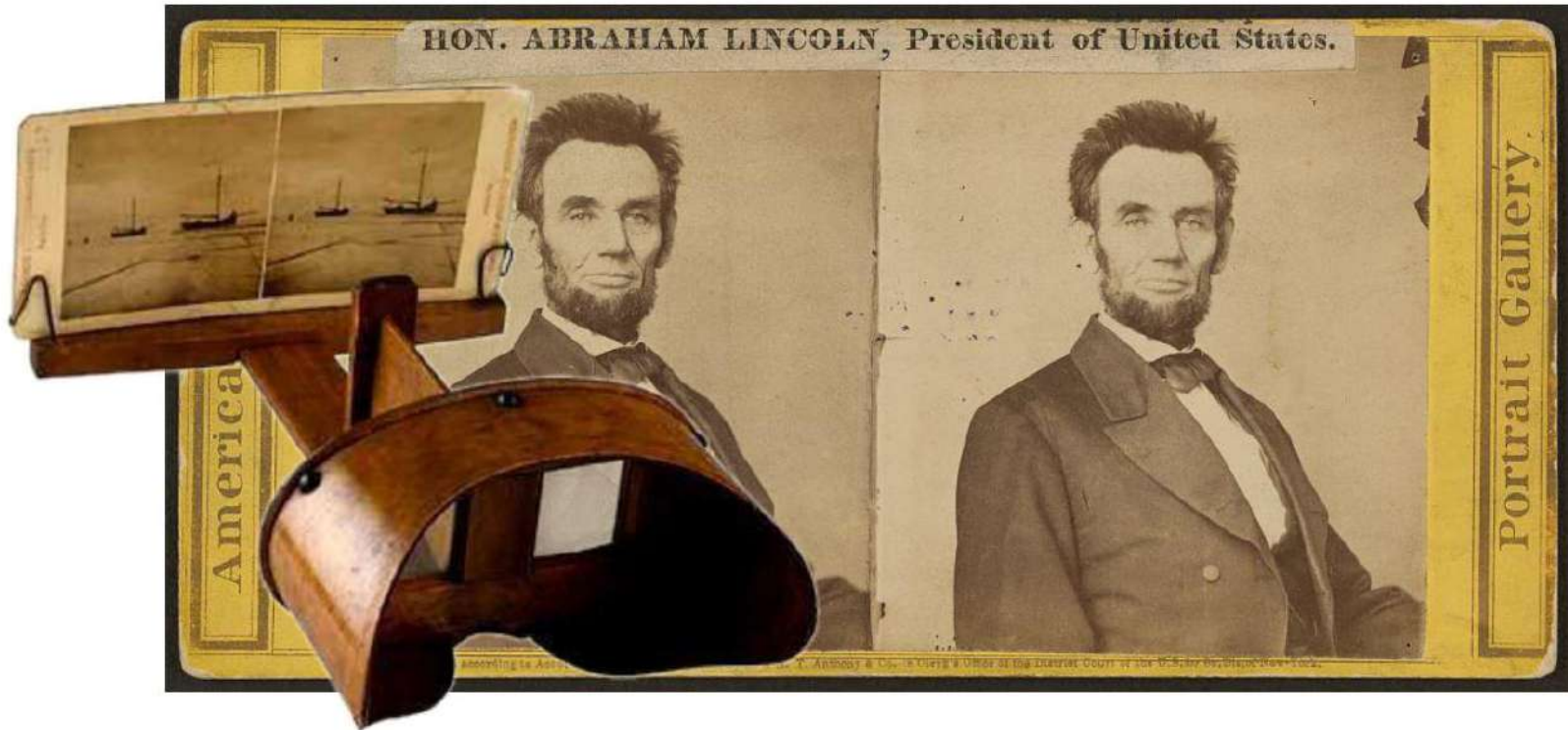
Gradient-Based Learning Applied to Document Recognition

YANN LECUN, MEMBER, IEEE, IÉON BOTTOU, YOSHUA BENGIO, AND PATRICK HAFFNER



Human visual system

Applications: Stereoscopic Displays

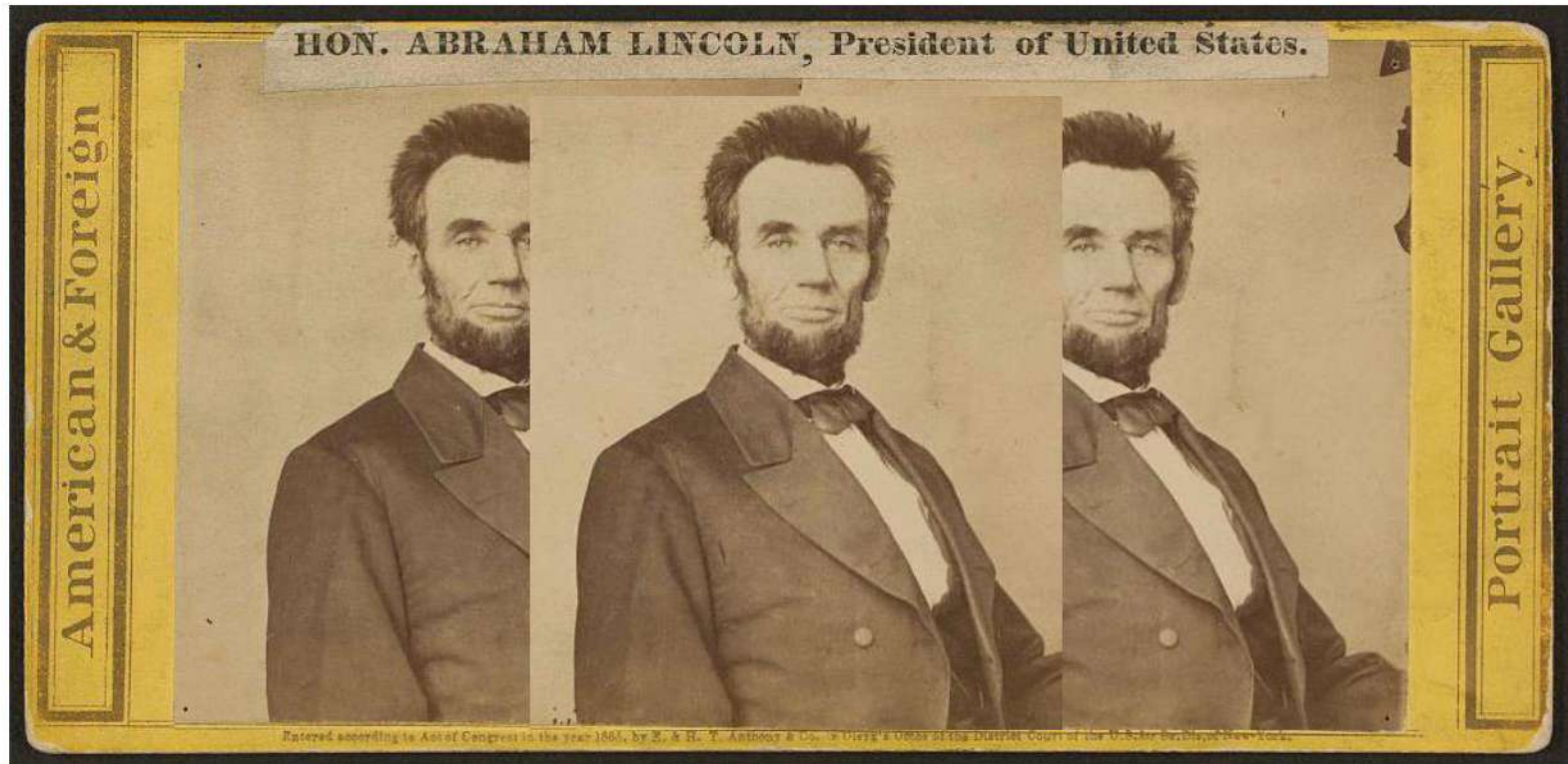


Charles Wheatstone., 1841. Stereoscope.

Walker, Lewis E., 1865. Hon. Abraham Lincoln, President of the United States. Library of Congress

Human visual system

Applications: Stereoscopic Displays



Human visual system

Applications: Stereoscopic Displays



Charles Wheatstone 1838



176 years later



stereoscopic displays

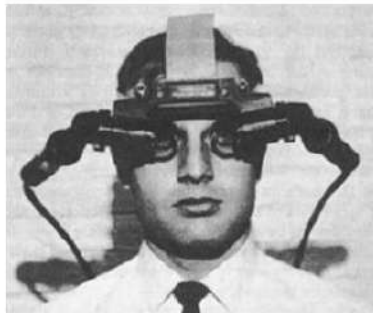
Human visual system

Applications: Virtual Reality

Stereoscopes
Wheatstone, Brewster, ...



VR, AR,
Ivan Sutherland



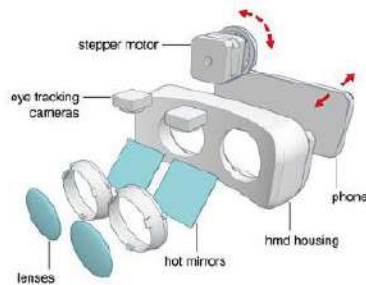
VR explosion
Meta, Apple, Sony, ...



Human visual system

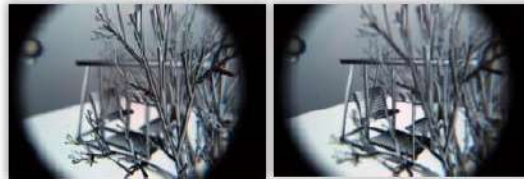
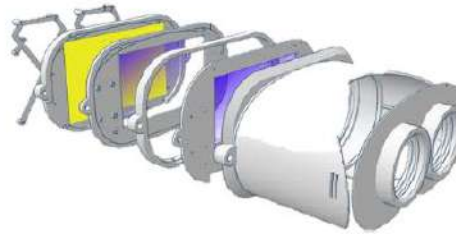
Applications: Virtual Reality

Gaze-contingent Focus Displays



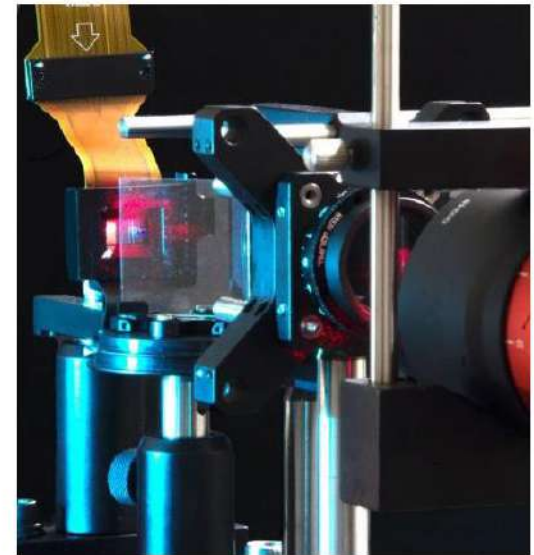
Konrad et al., SIGCHI 2016;
Padmanaban et al., PNAS 2017

Near-eye Light Field Displays



Huang et al., SIGGRAPH 2015;
Wetzstein et al., SIGGRAPH 2011, 2012

Holographic Displays

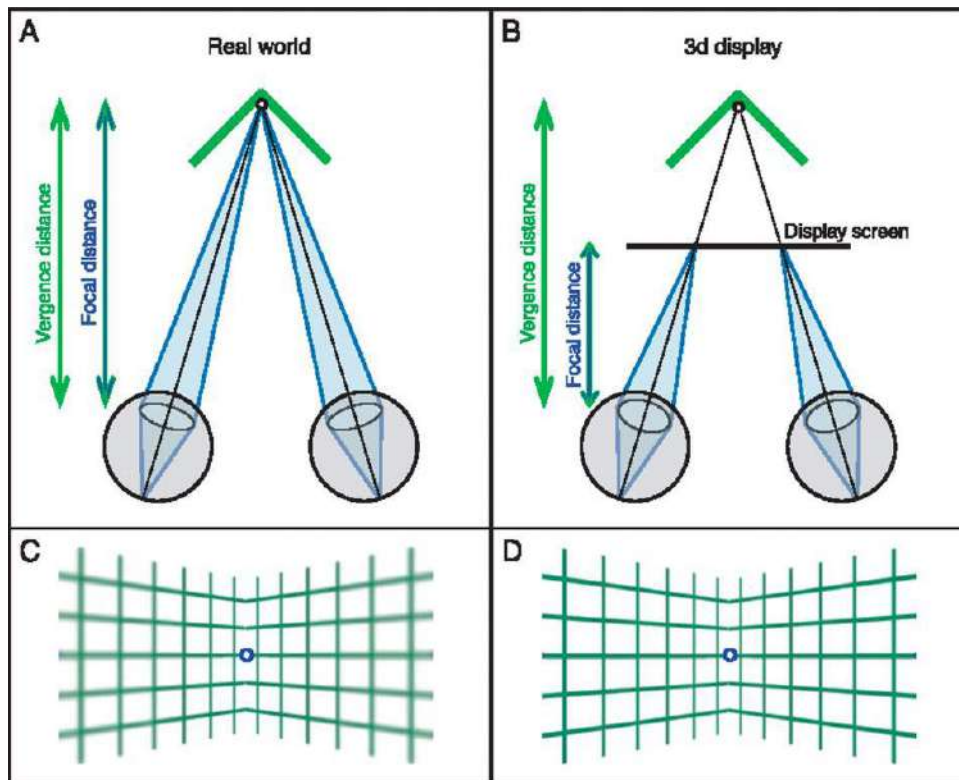


Padmanaban et al., SIGGRAPH Asia 2019
Peng et al., SIGGRAPH Asia 2020
Choi et al., Optica 2021

Human visual system

Applications: displays

Vergence-Accommodation Conflict



Effects:

- visual discomfort
- visual fatigue
- nausea
- diplopic vision
- eyestrain
- compromised image quality
- pathologies in developing visual system

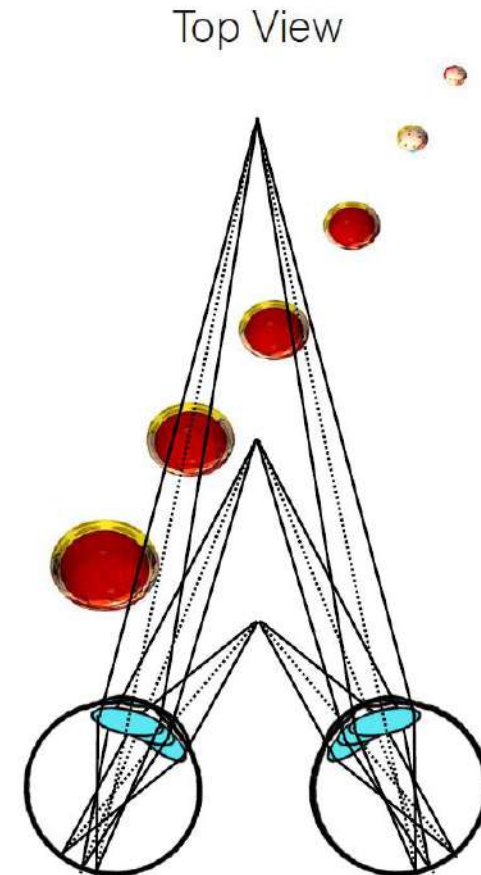
Human visual system

Applications: displays



Real World:

Vergence & Accommodation **Match!**



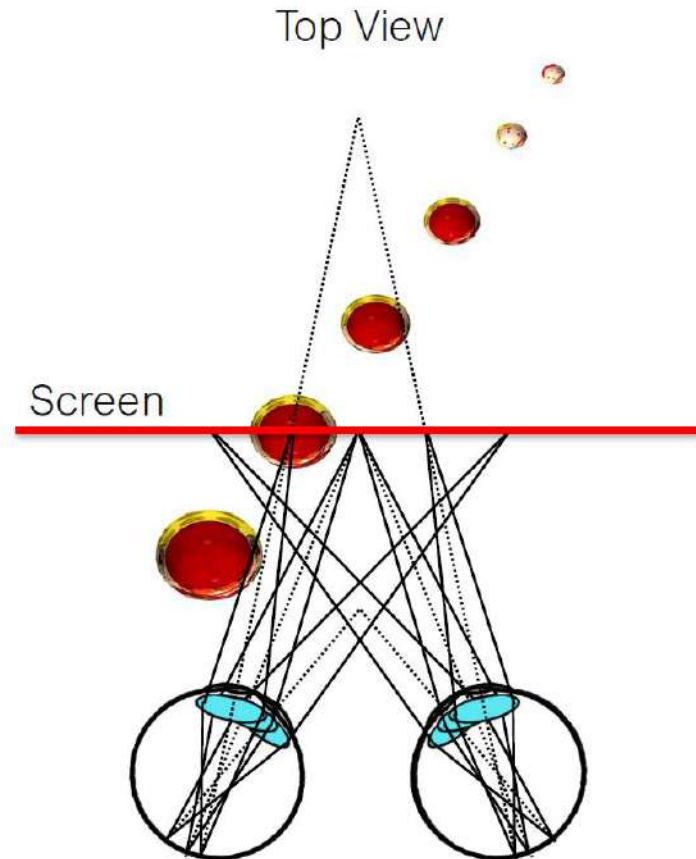
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Stereo Displays Today:

Vergence-Accommodation **Mismatch!**

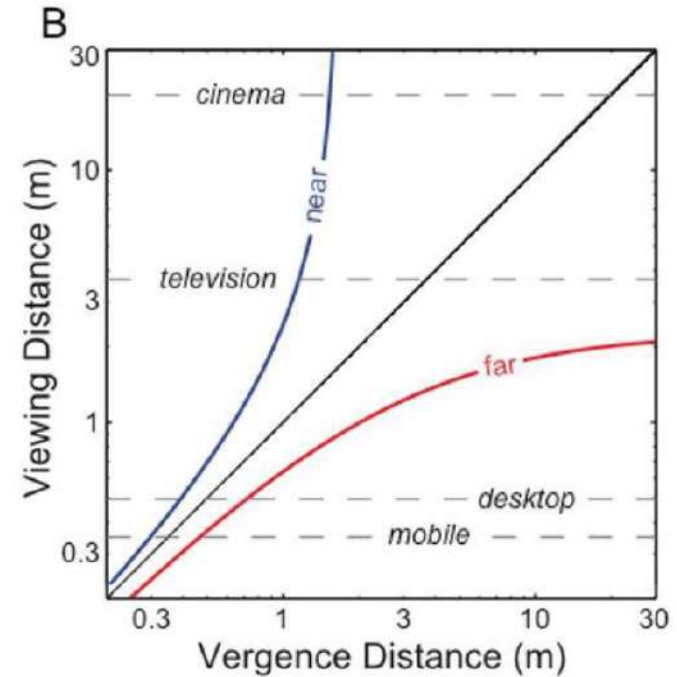
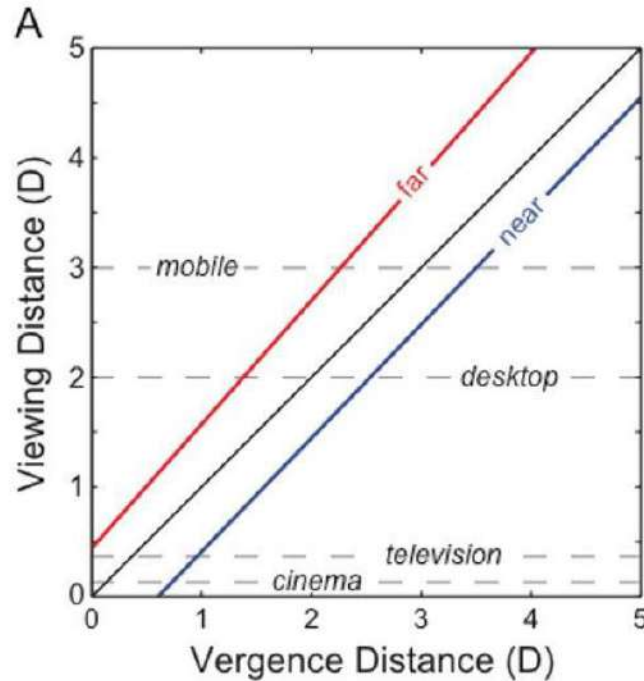


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Applications: displays

Zone of Comfort

Shibata et al, 2011, Journal of Vision



References

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- Computational Photography SIGGRAPH Course (Raskar & Tumblin)
- Digital and Computational Photography (Durand & Freeman, MIT)
- Computational Photography (Efros, CMU)
- Computational Photography (Gkioulekas, CMU)
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- Computational Photography (Fergus, NYU)
- Computational Imaging (Dragotti, Imperial College)
- Computer Vision (Seitz & Szeliski, UWashington)
- Introduction to Visual Computing and Visual Modeling (Kutulakos, UToronto)